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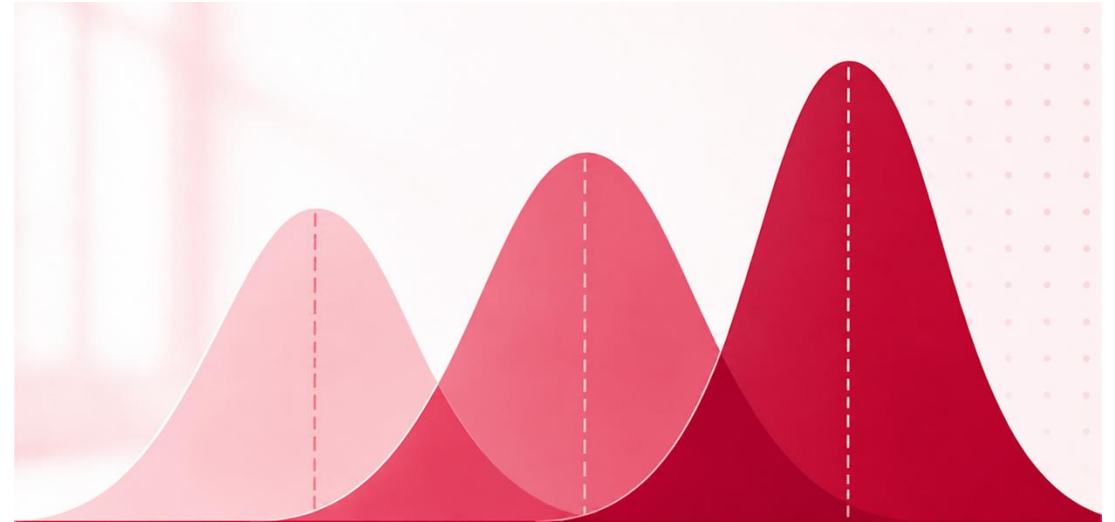
Introduction to Bayesian Inference

A Hands-on Workshop for Early-Career Researchers

Day 1

Julius M. Pfadt & Henrik R. Godmann

17-18 September 2026



Contents – Day 1

1. Introduction
2. Bayesian basics
3. Bayesian estimation
4. Bayesian testing
5. Bayesian benefits
6. JASP with real data
7. JASP AI
8. Conclusions

Contents – Day 2

1. Bayesian multi-model inference
2. Mixed-effects regression in R with *brms*
3. How to conduct and report a Bayesian analysis
4. Bring your own data
5. Beyond the workshop
6. Conclusions

Introduction – Who are we?

Julius M. Pfadt

- First contact with Bayesian statistics in 2017 at an internship at the University of Amsterdam (UvA) supervised by Eric-Jan Wagenmakers: Bayesian version of Cronbach's alpha
- PhD at Ulm University: (Bayesian) reliability estimation
- Post-doc at UvA (WB-fellowship): Bayesian model averaging for SEM
- Post-doc at UvA: Bayesian model comparison, large scale open science tracking



Introduction – Who are we?

Henrik R. Godmann

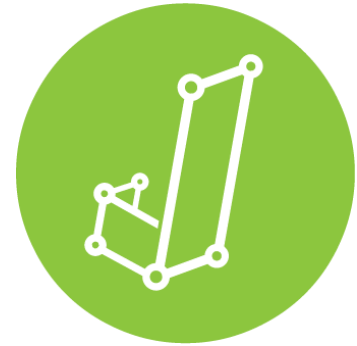
- First contact with Bayesian statistics in 2021 in my first master's course at the University of Amsterdam (UvA) led by Eric-Jan Wagenmakers
- TA / RA in Bayesian statistics with Eric-Jan Wagenmakers
- PhD Candidate in Bayesian time series and model averaging with Eric-Jan Wagenmakers



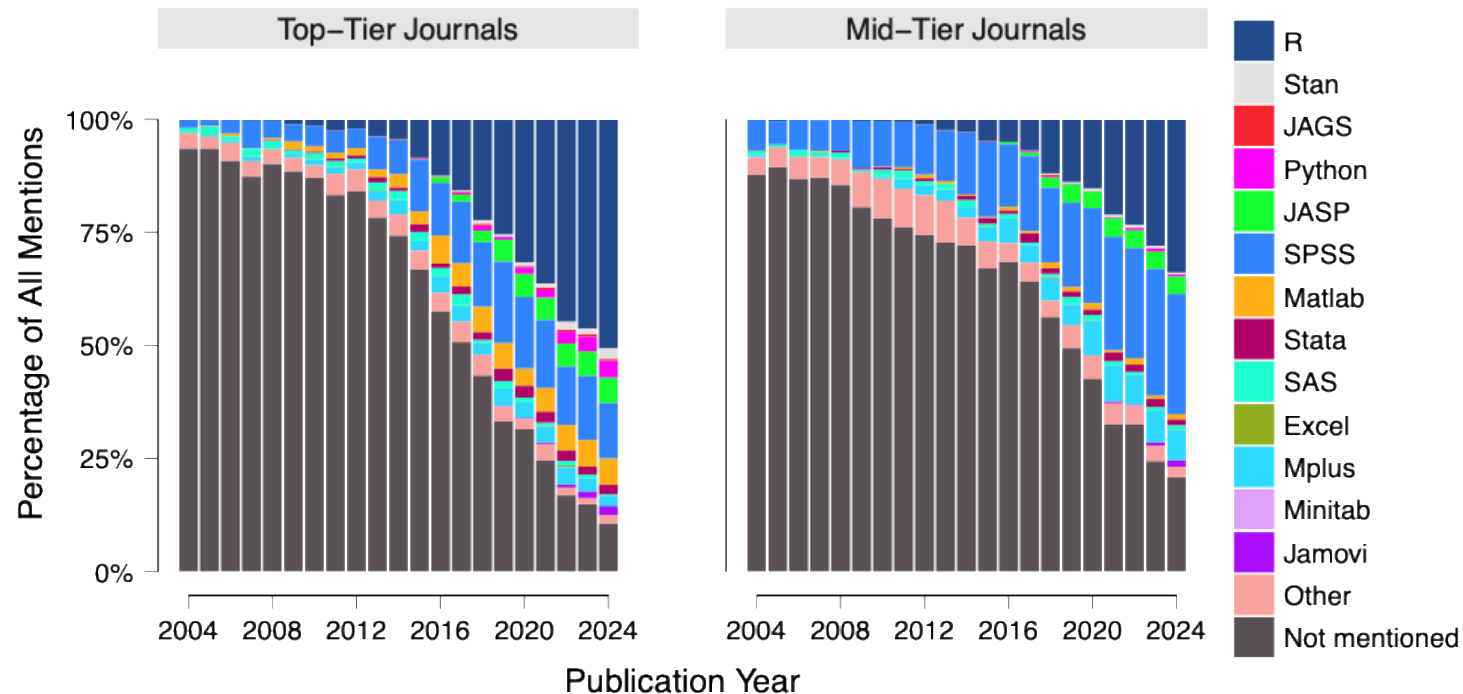
Introduction – Who are you?

- What is your research field?
- What methods are you using?
- What software are you using?
- Have you heard about Bayesian inference? If so, what have you heard?
- Have you used Bayesian methods? If so, what was your experience? If not, why not?

Introduction – Software: JASP



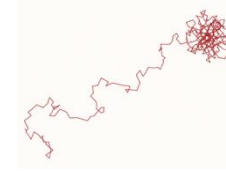
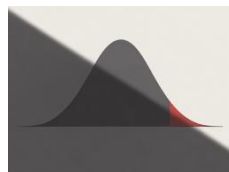
- Started in 2012 by EJ Wagenmakers at the UvA as a Bayesian alternative to SPSS
- Open source and free, no user data is collected
- Largest assortment of Bayesian analyses in the social sciences
- More frequentist than Bayesian analyses
- Now:



Pfadt et al. (2026)

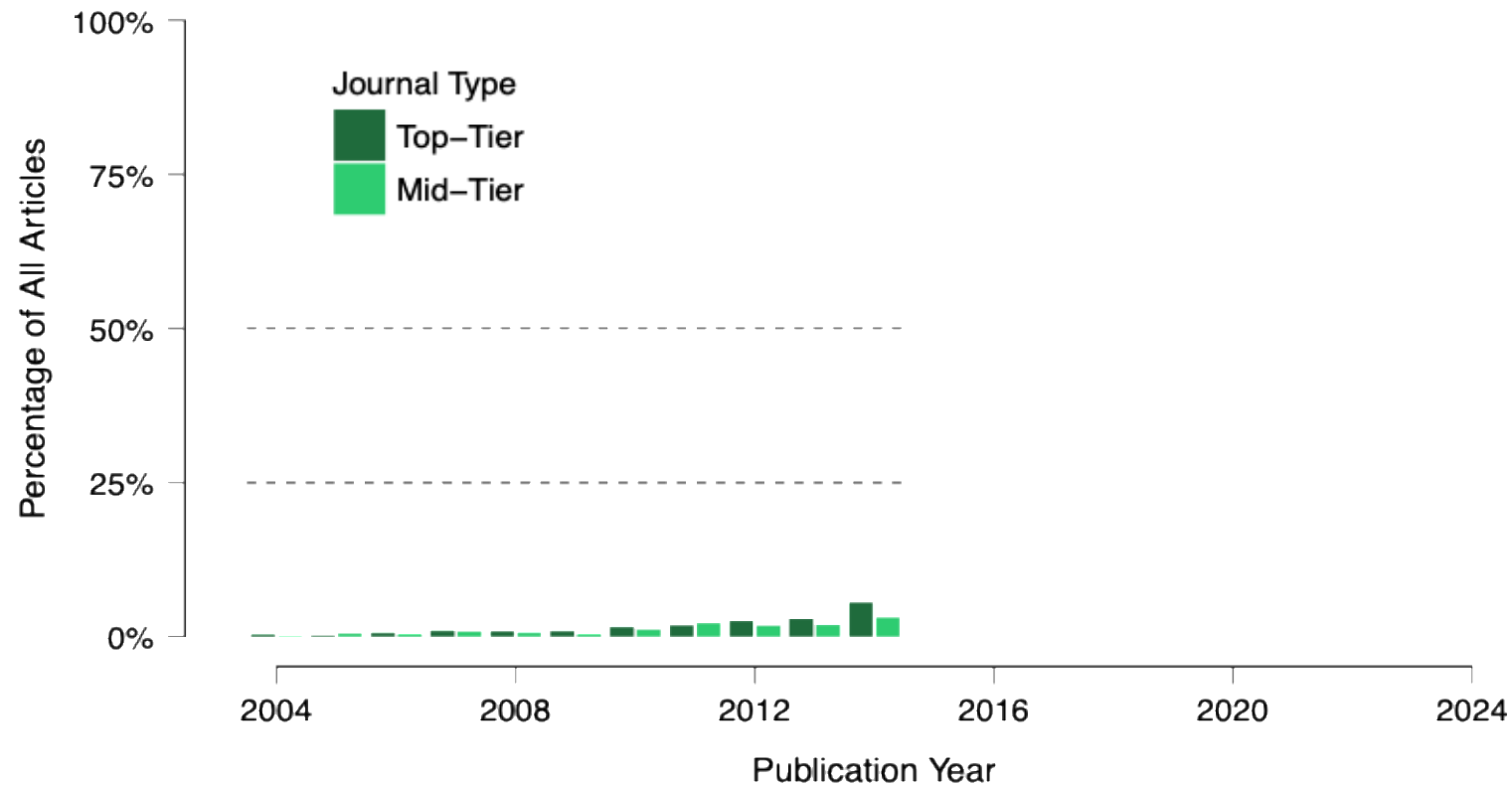
Introduction – History of Bayesian statistics

- 1763: Thomas Bayes works out conditional probability; Richard Price presents Bayes' work to the Royal Society
- 1774-1782: Laplace derives the same result but develops it into a working method for the sciences
- Early 20th century: Fisher, Neyman, and Pearson build frequentist framework
- 1939: Jeffreys publishes *Theory of Probability*; default priors, Bayes factor
- 1990s – Computational leap: MCMC makes posteriors computable, Bayesian computation becomes feasible
- 2011 – Replication crisis: Problems with p-values are exposed



Introduction – Today

How many people do you think use Bayesian statistics these days (Pfadt et al., 2026)?



Introduction – Goals

You shall not:

1. Unlearn everything you know about statistics
2. Become die-hard Bayesians
3. See the prior distribution as your enemy
4. Use SPSS



You shall:

1. Learn what a posterior distribution represents and how prior, likelihood, and data combine to produce it
2. Understand how a Bayes factor is quantifying evidence for one hypothesis vs another
3. Know how Bayesian inference differs from frequentist inference
4. Choose priors and assess how sensitive their conclusions are to that choice
5. Fit and interpret standard Bayesian analyses in JASP
6. Be able to check whether a model has converged and whether it describes the data adequately
7. Report a Bayesian analysis in a form suitable for publication
8. Know which tools to turn to for models beyond the scope of the workshop



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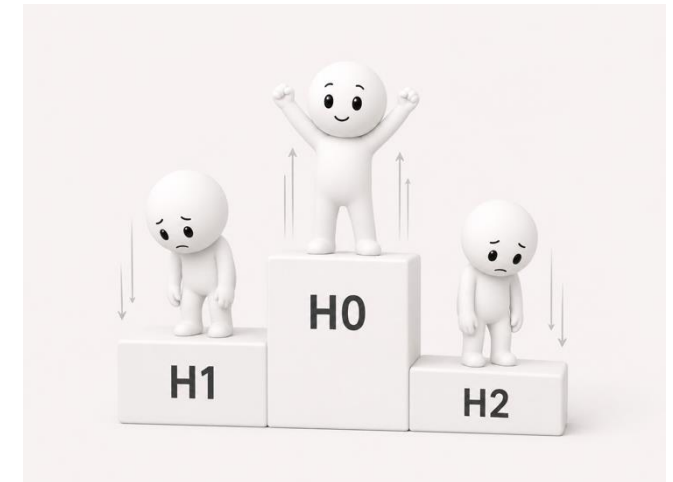
Bayesian basics – In a nutshell



You can state how confident you are.
In Bayesian inference, uncertainty or degree of belief is quantified by probability.



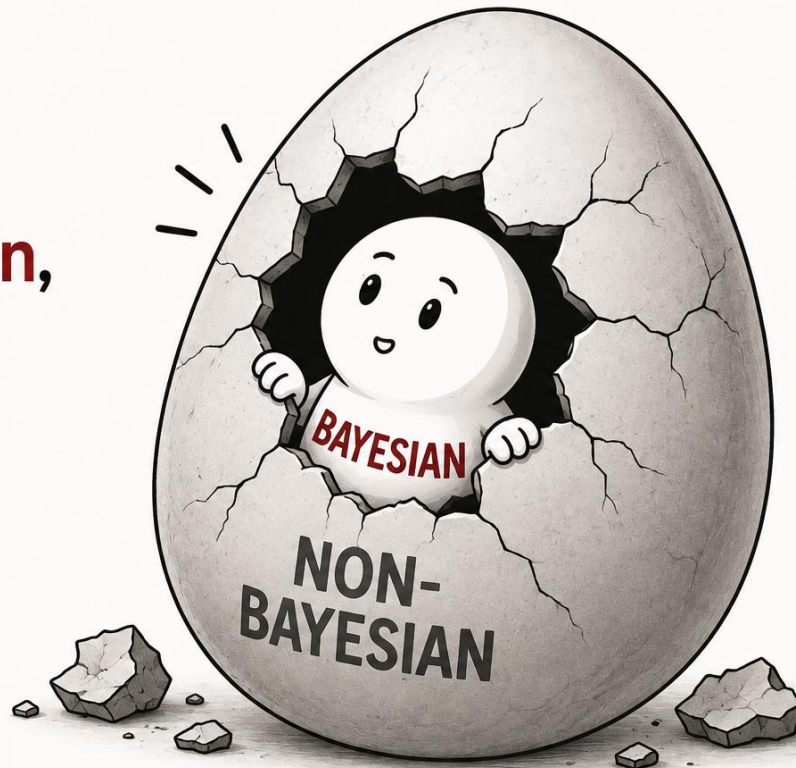
You can build on what you already know.
Prior uncertainty is updated by means of the data to yield posterior uncertainty.



You can find evidence for the absence of an effect.
Hypotheses that predict the data well enjoy a boost in credibility, whereas hypotheses that predict the data poorly suffer a decline.

Bayesian basics

Inside every
non-Bayesian,
there is a
Bayesian
struggling
to get out.



Dennis V. Lindley

Bayesian basics – Bayesian thinking



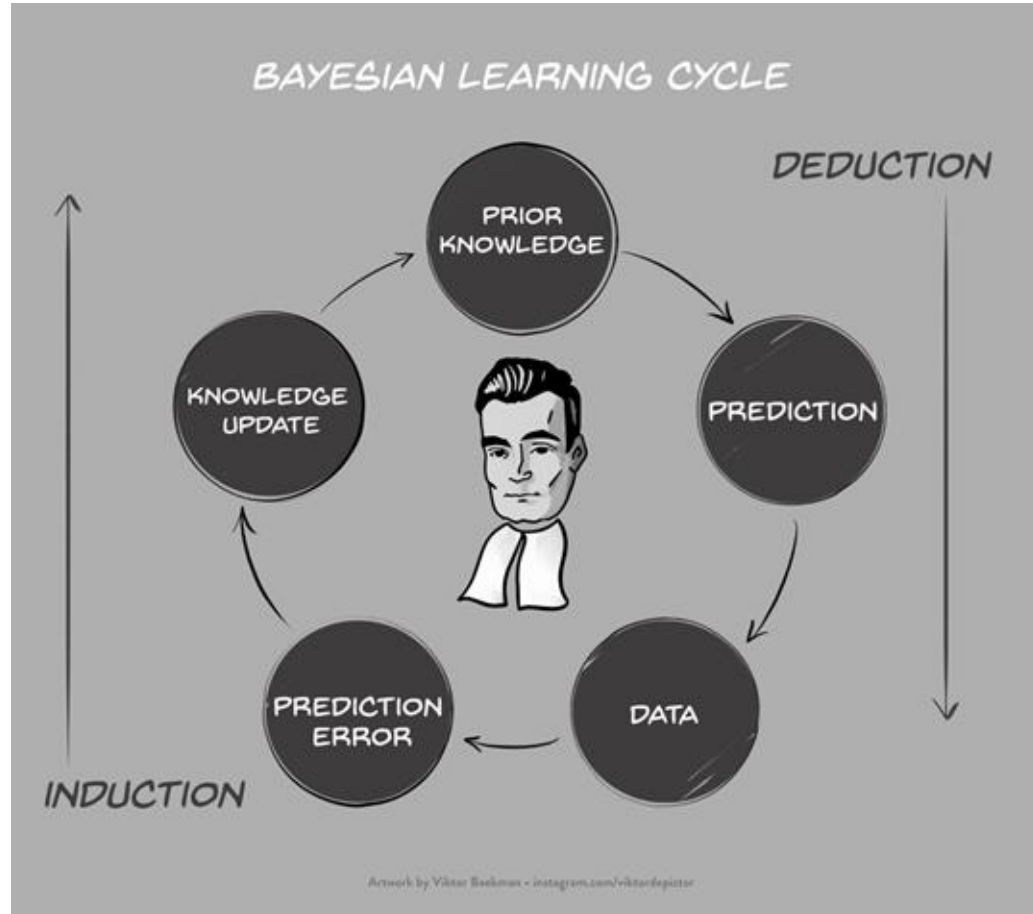
- Assume there are four dice
- You see the following outcomes: 2, 3, 3, 3, 1
- What die do you think generated these outcomes?

Bayesian basics – Bayesian thinking



- You saw: 2, 3, 3, 3, 1
- The next outcome is 5
- What die do you think generated these outcomes?
- You saw: 2, 3, 3, 3, 1, 5
- The next outcome is 12
- What die do you think generated these outcomes?
- Why would you not pick D60?

Bayesian basics – Learning cycle



- We pick a die for which we have prior knowledge
- We know what data this die would produce
- We observe outcomes
- Some outcomes may not agree with our prior beliefs
- We update our prior beliefs
- We pick another die
- ...

Bayesian basics – Frequentist probability

probability is the
long-run relative
frequency of an
event under
repeated trials



Bayesian basics – Probability

Can you assign a (frequentist) probability to the following statements?

- A coin lands heads on the next flip
- It rains in Giessen tomorrow
- The train will be late
- My intervention has an effect

Bayesian basics – Bayesian probability

probability is a
quantified
degree of belief



Bayesian basics – Probability

- I have a coin in my pocket
- Is it fair?



I do not know
If the coin were fair,
many sets of 100 flips
would mostly
produce close to 50
heads.

When the coin comes up heads 60
times in 100 flips, a fair coin would
produce a result this extreme about
6% of the time. That is not rare
enough to reject fairness — so I
have no reason to reject it.
I observed one set of 100 flips. But I
judged it against the distribution of
all the sets I did not run.

Bayesian basics – Probability

- I have a coin in my pocket
- Is it fair?

I think it is probably close to fair. Flip it once, twice, or a hundred times, and each flip updates what I believe about the coin's bias.



Before flipping, I thought the coin was probably close to fair. After seeing 60 heads in 100 flips, there is a 97% probability that this coin favours heads.



Bayesian basics – Generating knowledge



State of the World

Predictions

Data

"Given the state of the world what can we conclude about the data" -> $p(\text{data} \mid M)$



State of the World

Learning

Data

"Given the observed data what can we conclude about the state of the world" -> $p(M \mid \text{data})$

Bayesian basics – Bayes' Theorem

- We want to learn about the state of the world given the data: $p(M \mid data)$
- For researchers, the state of the world can be anything:
 - Hypothesis
 - Model
 - Parameter
- How do we obtain $p(M \mid data)$?
- Conditional probability (Bayes' Theorem):

$$p(A \mid B) = \frac{p(B \mid A) \cdot p(A)}{p(B)}$$

Bayesian basics – Bayes' Theorem

$$p(A | B) = \frac{p(B | A) \cdot p(A)}{p(B)}$$

$$p(M | \text{data}) = \frac{p(\text{data} | M) \cdot p(M)}{p(\text{data})}$$

- $p(M|\text{data})$: What we believe about the state of the world; the **posterior** probability
- $p(\text{data}|M)$: How well this model and these parameter values predict the data; the **likelihood**
- $p(M)$: What we believe before seeing the data; the **prior**
- $p(\text{data})$: The probability of the data, over all candidate models and parameter values; a normalizing constant
- After the break we look at how to obtain the posterior of a parameter
- If we manage to estimate all parts of the righthand side of the theorem we can obtain the probability of a hypothesis, model, parameter given the data



Break time

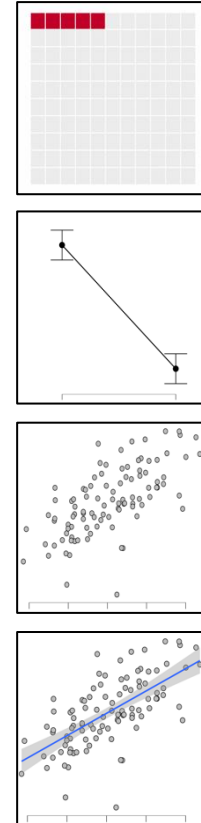


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Bayesian estimation – Introduction

- A proportion: What's the proportion of people with Long-Covid in Germany?
- A mean difference: Average symptom score before and after a treatment
- A correlation: The relationship between income and happiness
- A regression coefficient: How much next-day mood changes per unit of sleep quality
- We estimate a parameter, θ
- We do not only want to estimate a single value, for instance the mean of θ , but to quantify uncertainty about θ after seeing the data, that is, which values of θ are more or less plausible.

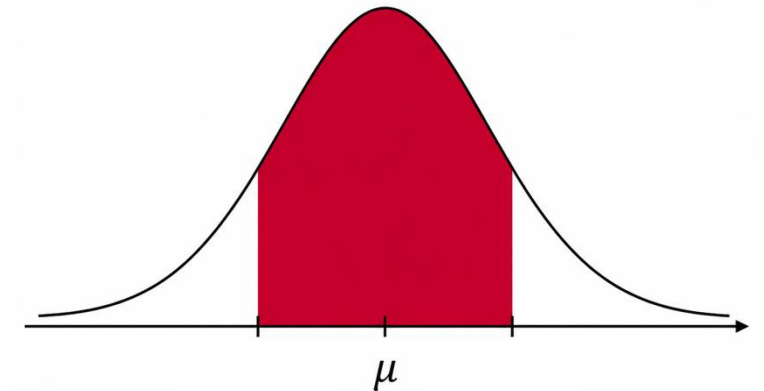


Bayesian estimation – Bayes theorem

- We assume a fixed model, then:

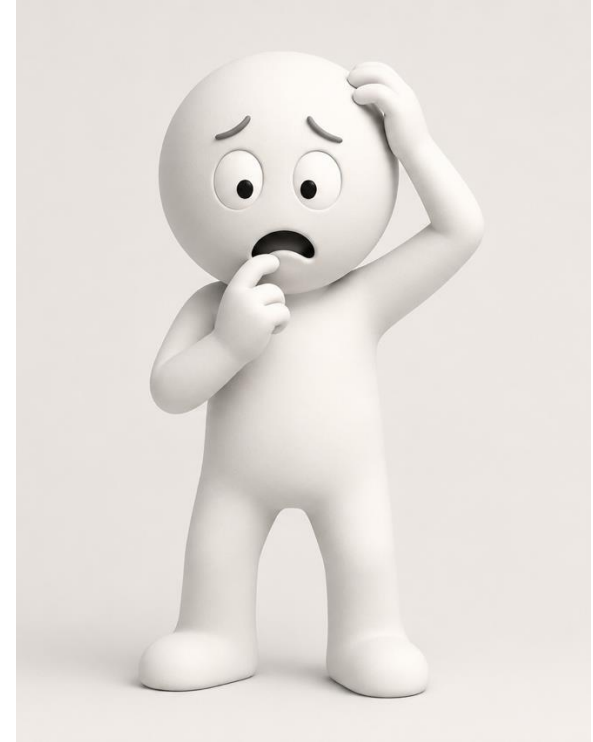
$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) \cdot p(\theta)}{p(\text{data})} = p(\theta) \frac{p(\text{data} \mid \theta)}{p(\text{data})}$$

- $\frac{p(\text{data} \mid \theta)}{p(\text{data})}$: predictive updating factor
- $p(\text{data})$ normalizes (rescales) the posterior; because it is the same number for every value of θ it does not affect which values of θ are plausible relative to each other, so we do not need to compute it
- $p(\theta \mid \text{data}) \propto p(\text{data} \mid \theta) \cdot p(\theta)$
- We express both the prior and the posterior as *probability distributions* because θ is unknown, and we use probability to describe uncertainty
- Posterior probability distribution can give us:
 - The most plausible parameter value
 - The probability that the parameter exceeds any value X
 - The interval that contains the parameter value with, e.g., 95% probability
- Example



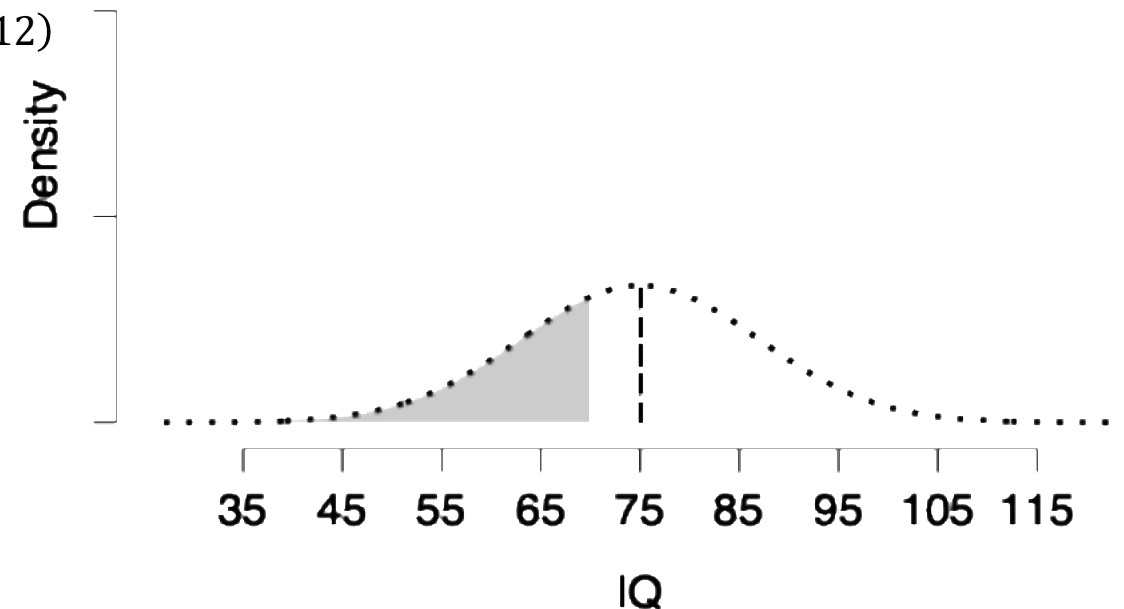
Bayesian estimation – “Florida Bob”

- “Florida Bob” killed his wife and may get the death penalty.
- The defense claims that Bob is intellectually disabled: Bob’s IQ is lower than 70.
- The judge orders three new IQ tests. The resulting scores: {73, 67, 79}.
- What can we conclude?



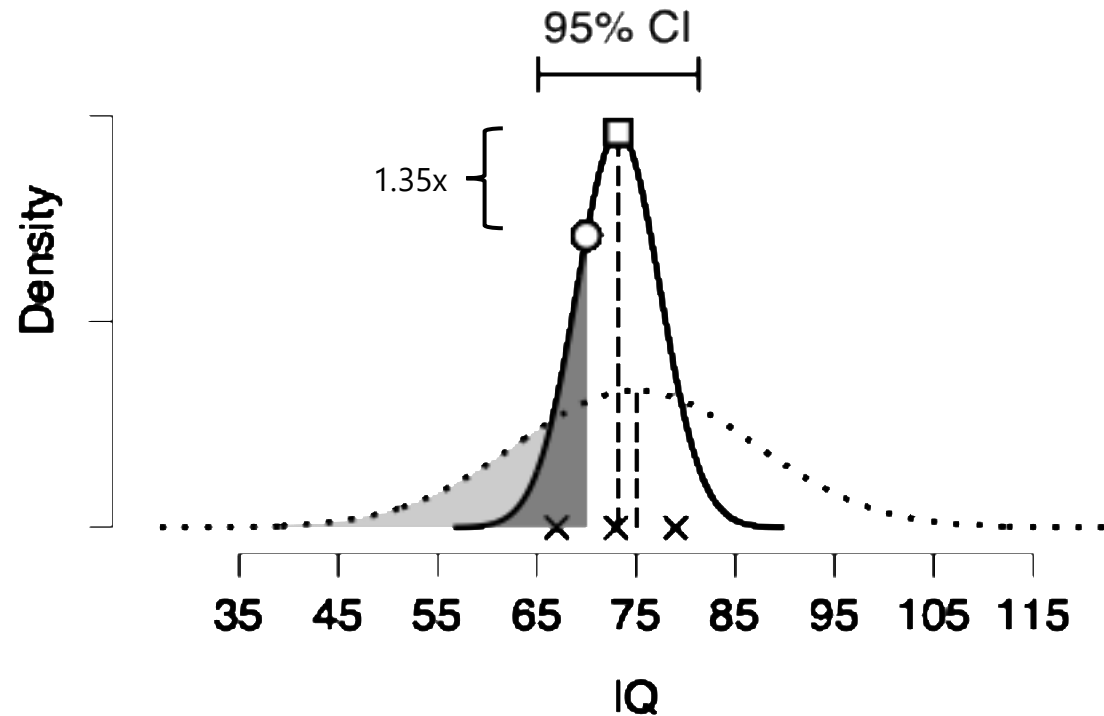
Bayesian estimation – “Florida Bob”

- The literature shows that, as a group, inmates classified as “intellectually disabled” have an IQ of $\text{Normal}(75, SD = 12)$.
- We use this knowledge to quantify our uncertainty about Bob's IQ
- Our prior for Bob's mean IQ is $p(\mu) = \text{Normal}(75, SD = 12)$
- Prior: $p(IQ < 70) = 0.34$



Bayesian estimation – “Florida Bob”

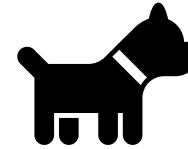
- We observe three IQ-tests with scores 73, 67, 79
- The likelihood: $p(\text{data} \mid \mu) = \text{Normal}(\text{data} \mid \mu, \sigma)$
- Posterior: $\text{Normal}(73.24, SD = 4.16)$
- $p(IQ < 70 \mid \text{data}) = 0.22$
- 95% credible interval: [65.09, 81.39]
- Bob having an IQ of 73.24 is 1.35 times as probable as the value of 70



Bayesian estimation – Binomial example

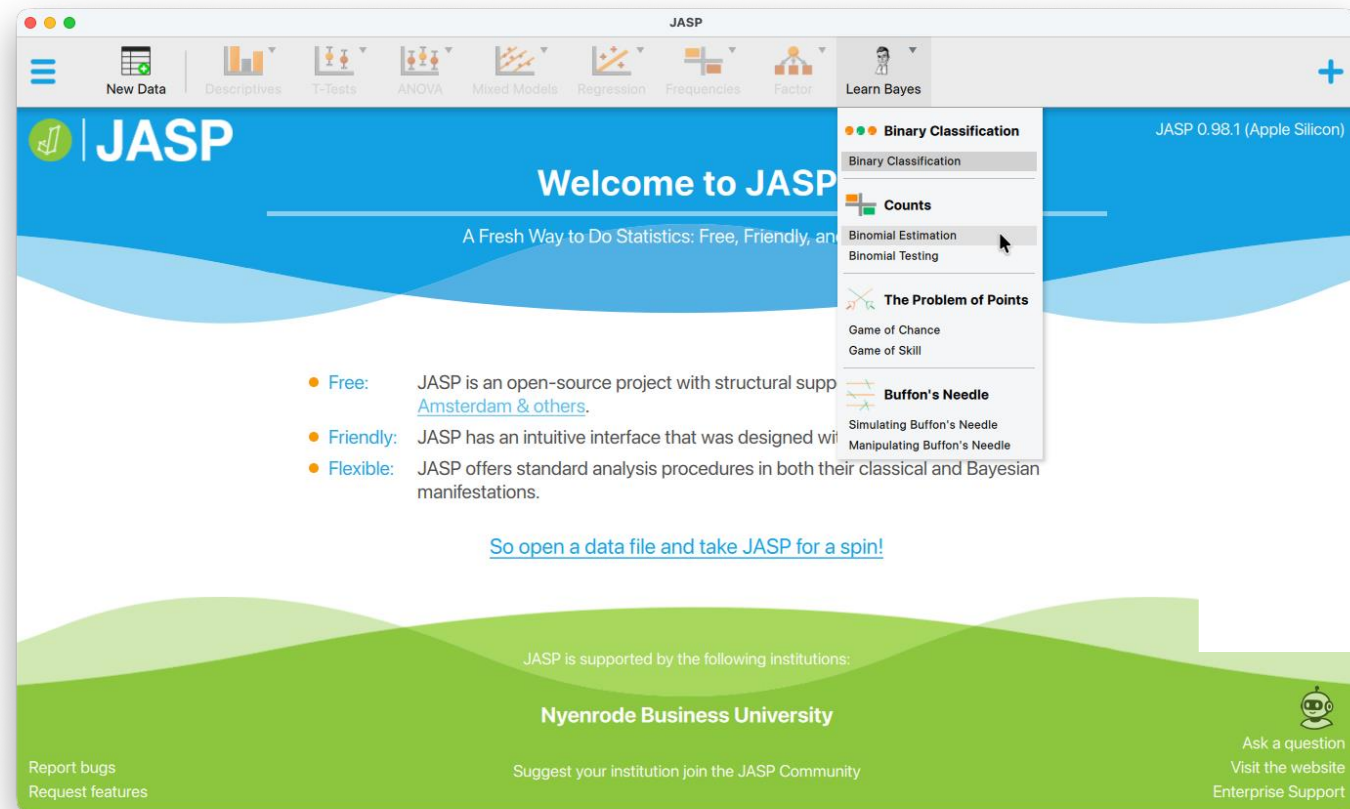


or



- Let's estimate the latent proportion, θ , of people (you) who prefer cats over dogs.
 - We will ask you to indicate whether you are a "cat person" or a "dog person".
 - Three ingredients: prior, likelihood, and posterior
 - **Prior:** Belief about the proportion of cat lovers before asking anyone -> Beta distribution, $\theta \sim \text{Beta}(a, b)$
 - **Likelihood:** Probability of the observed count of cat lovers in the room:
 - Binomial, $p(k | \theta) = \text{Binomial}(k | n, \theta)$, where k is the successes (cat-lovers) out of n trials
- $$p(k | \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$
- **Posterior:** Belief about the proportion of cat lovers after observing the counts:
 - Beta distribution: $p(\theta | k) = \text{Beta}(a + k, b + n - k)$

Bayesian estimation – Binomial example in JASP



Bayesian estimation – Beyond

- Usually, the prior and posterior are not as well behaved
- Models have many parameters and their multivariate distributions are not analytically tractable
- In that case we approximate the posterior with Markov chain Monte Carlo methods (MCMC)
- More on that tomorrow

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Bayesian testing – Introduction

- Is there any effect of this treatment at all?
 - Is this coin fair?
 - Do these two groups differ?
 - Is this correlation different from zero?
 - Does adding this predictor improve the model?
-
- In testing we are interested in at least two competing models/hypotheses, e.g., H_0 and H_1
 - Which hypothesis is better supported by the data?
 - Which hypothesis predicted the data best?
 - The ratio of predictive performance of two hypotheses is known as the **Bayes factor**



Bayesian testing – Bayes' theorem

$$p(H \mid \text{data}) = \frac{p(\text{data} \mid H) \cdot p(H)}{p(\text{data})}$$

- $p(\text{data})$: the probability of the data across all hypotheses under consideration (and their parameters):

$$p(\text{data}) = \sum_i p(\text{data} \mid H_i) p(H_i)$$

- We transform to:

$$\underbrace{\frac{p(H \mid \text{data})}{p(H)}}_{\text{Support}} = \underbrace{\frac{p(\text{data} \mid H)}{p(\text{data})}}_{\text{Predictive success}}$$

- **Support**: How much our belief in the hypothesis changed
- **Predictive success**: How well the hypothesis predicted the data *compared to* the average across all hypotheses under consideration

Bayesian testing – Bayes' theorem

- We want to compare two hypotheses, H_1 and H_0 :

$$p(H_1|\text{data}) = p(H_1) \frac{p(\text{data}|H_1)}{p(\text{data})}, \quad p(H_0|\text{data}) = p(H_0) \frac{p(\text{data}|H_0)}{p(\text{data})}$$

- We compare each hypothesis' predictive success relative to the other:

$$\frac{p(\text{data}|H_1)}{p(\text{data})} / \frac{p(\text{data}|H_0)}{p(\text{data})} = \frac{p(\text{data}|H_1)}{p(\text{data}|H_0)}$$

- And:

$$\underbrace{\frac{p(H_1|\text{data})}{p(H_0|\text{data})}}_{\text{Posterior odds}} = \underbrace{\frac{p(H_1)}{p(H_0)}}_{\text{Prior odds}} \times \underbrace{\frac{p(\text{data}|H_1)}{p(\text{data}|H_0)}}_{\text{Bayes factor}}$$

- **Prior odds:** Our belief about how much more probable H_1 is than H_0 before seeing the data
- **Posterior odds:** How much more probably H_1 is than H_0 after seeing the data

Bayesian testing – Bayes factor

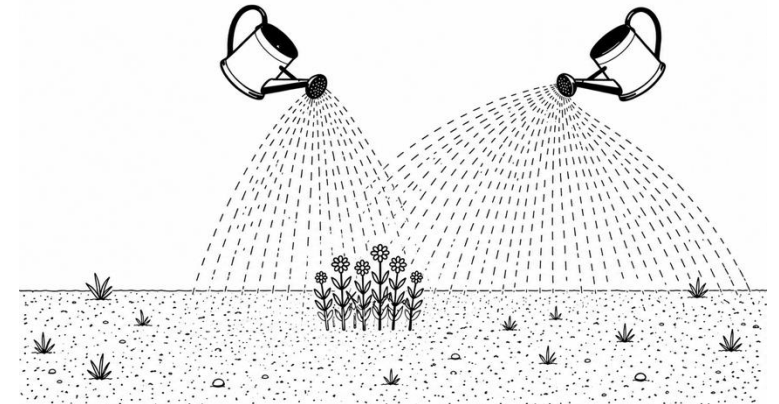
- BF_{10} : How much more probable the data are under H_1 than H_0
 - $BF_{01} = 1/BF_{10}$
 - $BF_{10} > 1$: The data are more likely under H_1 than under H_0
 - $BF_{10} < 1$: The data are more likely under H_0 than under H_1
 - $BF_{10} = 5$: The data are 5 times more likely under H_1 than under H_0
 - $BF_{10} = 1/5$: The data are 5 times more likely under H_0 than under H_1
-
- ➔ Relative measure of evidence: Evidence of an effect, Evidence of absence, absence of evidence
 - ➔ Evidence (the BF) should not be confused with belief (posterior probability); they are the same under one condition; which one?
 - ➔ The parameter prior distributions are important; vague prior distributions are a bad idea. Why?

Bayesian testing – Marginal likelihood

- The marginal likelihood:

$$p(\text{data} \mid H) = \int \underbrace{p(\text{data} \mid \theta, H)}_{\text{likelihood}} \underbrace{p(\theta \mid H)}_{\text{prior}} d\theta$$

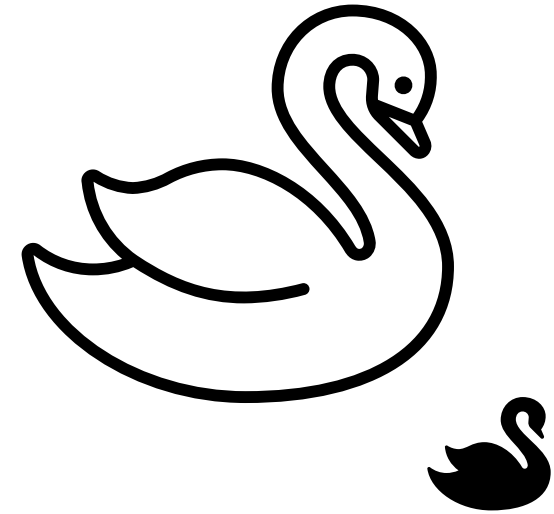
- The integral is an average over all the hypothesis's possible parameter values
- Product of likelihood and prior: If both densities are high at the same values of θ , the product is large and so is the integral
- If the prior is wide: its density is spread thinly across many values of θ , so even where the likelihood is high, the prior contributes only a small weight there, thus, the product stays low almost everywhere, and the integral is small.
- If the prior is narrow: its density is concentrated on a few values of θ . If that concentration lands where the likelihood is high, the product is large exactly there and the integral is large. If it lands where the likelihood is low, the integral is small too



Bayesian testing – Example I: Binomial hypothesis test

Universal generalization:

- All X are Y
- All swans are white
- All zombies are hungry
- Every confirmatory instance increases our confidence in the general law
- We represent the hypotheses with priors:
 - $H_0: \theta = 1$ -> prior at point value
 - $H_1: \theta \neq 1$ -> prior: $\text{Beta}(a, b)$
- The beta-binomial model:
$$BF_{01} = \frac{p(\text{data} \mid H_0)}{p(\text{data} \mid H_1)} = n + 1$$



Bayesian testing – Example I: Binomial hypothesis test

The screenshot displays the JASP software interface, specifically the Binomial Testing module. The top toolbar includes icons for New Data, Descriptives, T-Tests, ANOVA, Mixed Models, Regression, Frequencies, Factor, and Learn Bayes. The main window is divided into two panes. The left pane, titled 'Binomial Testing', contains settings for the analysis: 'Introductory text' is unchecked, 'Color palette' is set to 'Colorblind', 'Data' is selected, 'Input Type' is 'Specify counts', and 'Count Data' shows 'Successes' and 'Failures' both at 0. The 'Hypothesis' section has a table with columns for Hypothesis, Prior prob., Distribution, Parameter (θ), and Truncation. Below this are expandable sections for 'Prior and Posterior', 'Prior Predictive Performance', 'Sequential Analysis', and 'Posterior Prediction'. The right pane, titled 'Results', shows a 'Binomial Test' section with 'Hypotheses' and a 'Testing Summary' table. A dropdown menu is open over the 'Learn Bayes' icon, listing various Bayesian models: 'Binary Classification', 'Counts', 'Binomial Estimation', 'Binomial Testing' (highlighted), 'The Problem of Points', 'Game of Chance', 'Game of Skill', 'Buffon's Needle', 'Simulating Buffon's Needle', and 'Manipulating Buffon's Needle'.

JASP*

New Data Descriptives T-Tests ANOVA Mixed Models Regression Frequencies Factor Learn Bayes

Binomial Testing

☐ Introductory text Color palette Colorblind

Data

Input Type

☐ Select variable ☒ Specify counts ☐ Enter sequence

Count Data

Successes 0

Failures 0

Hypothesis

Hypothesis	Prior prob.	Distribution	Parameter (θ)	Truncation

Prior and Posterior

Prior Predictive Performance

Sequential Analysis

Posterior Prediction

Results

Binomial Test

Hypotheses

Testing Summary

Hypothesis		

Binary Classification

Counts

Binomial Estimation

Binomial Testing

The Problem of Points

Game of Chance

Game of Skill

Buffon's Needle

Simulating Buffon's Needle

Manipulating Buffon's Needle

Bayesian testing – Example II: Binomial hypothesis test

- Does a coin tend to land on the same side it was flipped from?
- H_0 : the probability for same-side outcome is 50%
- H_1 : the probability for same-side outcome is not 50%
- 350,757 flips
- 178,079 same-side outcomes
- 172,678 other-side outcomes
- The beta-binomial model:
 - $H_0: \theta_0 = .5$
 - $H_1: \theta_0 \sim \text{Beta}(a, b)$

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<https://doi.org/10.1080/01621459.2025.2516210>



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Fair Coins Tend to Land on the Same Side They Started: Evidence from 350,757 Flips

František Bartoš^a, Alexandra Sarafoglou^a, Henrik R. Godmann^a, Amir Sahrani^a, David Klein Leunk^a, Pierre Y. Gui^b, David Voss^b, Kaleem Ullah^b, Malte Zoubek^{c,d}, Franziska Nippold^b, Frederik Aust^e, Felipe Fontana Vieira^{d,e,f}, Chris-Gabriel Islam^{g,h}, Anton J. Zoubek^g, Sara Shabani^h, Jonas Petterⁱ, Ingeborg B. Roosⁱ, Adam Finnemann^{j,k}, Aaron B. Lob^{b,k}, Madlen F. Hoffstadt^a, Jason Nak^b, Jill de Ron^b, Koen Derks^l, Karoline Huth^{a,m}, Sjoerd Terpstraⁿ, Thomas Bastelica^{o,p}, Magda Matetovici^{b,q}, Vincent L. Ott^b, Andreea S. Zetea^b, Katharina Karnbach^b, Michelle C. Donzallaz^r, Arne John^s, Roy M. Moore^b, Franziska Assion^r, Riet van Bork^a, Theresa E. Leidinger^b, Xiaochang Zhao^b, Adrian Karami Motaghi^b, Ting Pan^{b,j}, Hannah Armstrong^b, Tianqi Peng^{b,i}, Mara Bialas^{b,s}, Joyce Y.-C. Pang^t, Bohan Fu^b, Shujun Yang^b, Xiaoyi Lin^b, Dana Sleiffer^b, Miklos Bognar^r, Balazs Aczel^u, and Eric-Jan Wagenmakers^a

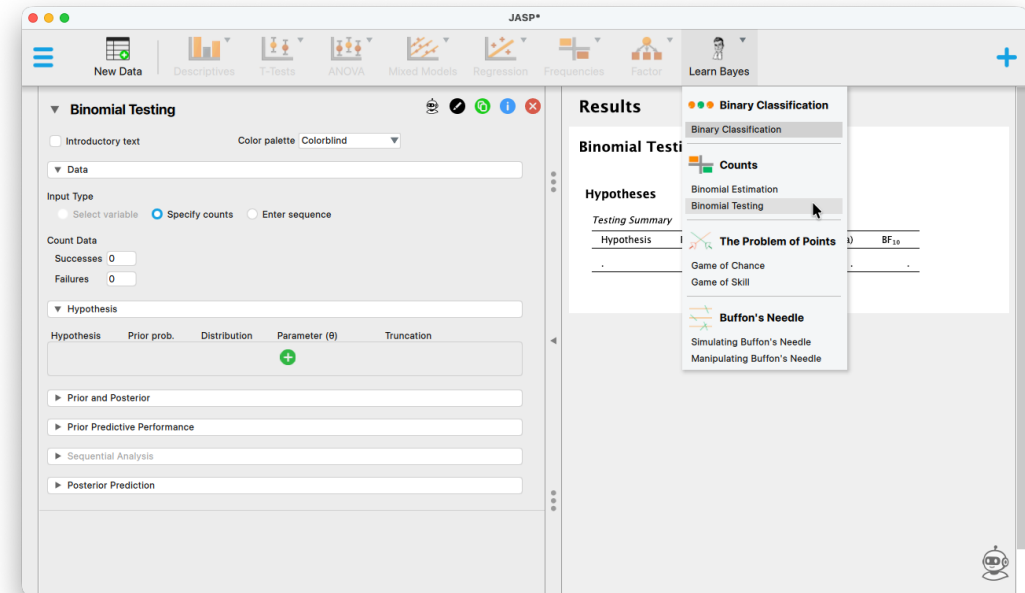
Bayesian testing – Example II: Binomial hypothesis test

In JASP: Try expressing different alternative hypotheses with beta distributions, visualize the prior distribution

- Same-side bias
- Heads-tails-bias

In the paper:

- Same-side bias: $\text{Beta}(5100, 4900)$
- Heads-tails-bias: $\text{Beta}(5000, 5000)$



Bayesian testing – Bayes factor interpretation

- Numeric:

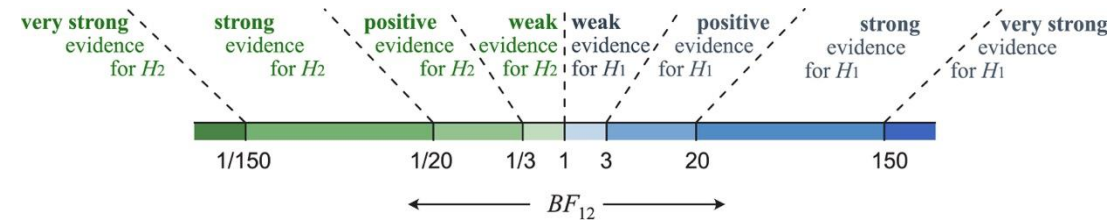


Figure from Mulder, Friel, & Leifeld (2024)

- Pizza and mozzarella:

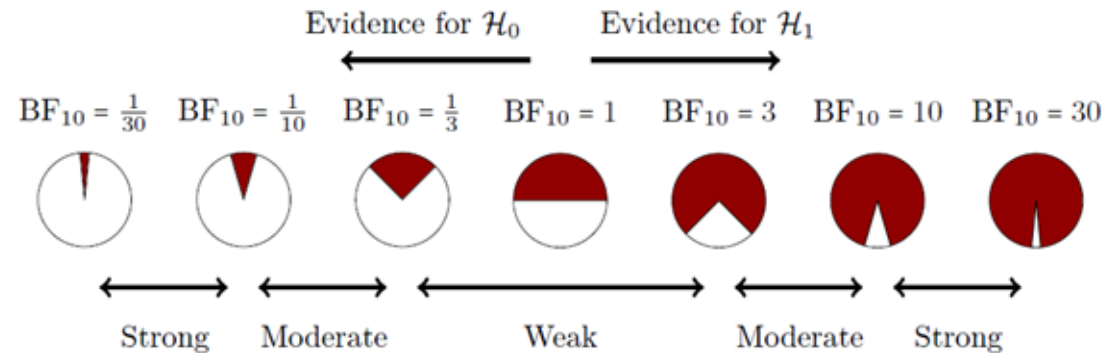


Figure from Van Doorn et al. (2021)

- Presenting the BF on the log scale:

$$\log(BF_{10}) = \log p(\text{data} | H_1) - \log p(\text{data} | H_0) = \ln(BF_{10})$$

- Symmetric: below 0 support for the alternative, above 0 support for the null
- Numerically stable
- Averaging

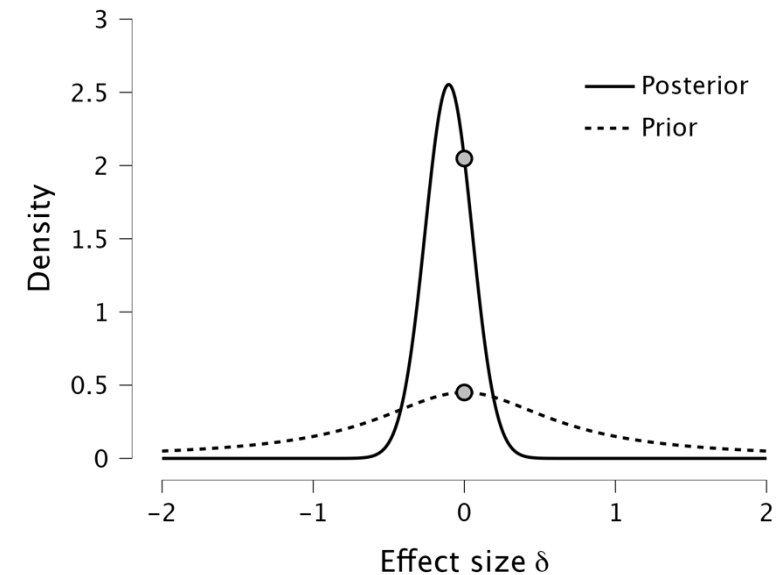
Bayesian testing – Beyond

- For the beta-binomial model the marginal likelihood for the BF is available analytically
- Unfortunately, that is an exception
- Approximating the marginal likelihood is often complicated

A shortcut is the **Savage-Dickey** density ratio:

$$BF_{01} = \frac{p(\theta_0 \mid \text{data}, H_1)}{p(\theta_0 \mid H_1)}$$

- Posterior density at the null value, divided by the prior density at the null value, under H_1
- Requires: Estimation of the prior and posterior densities
- Conditions:
 - Nested hypotheses
 - Point null
 - Independent priors



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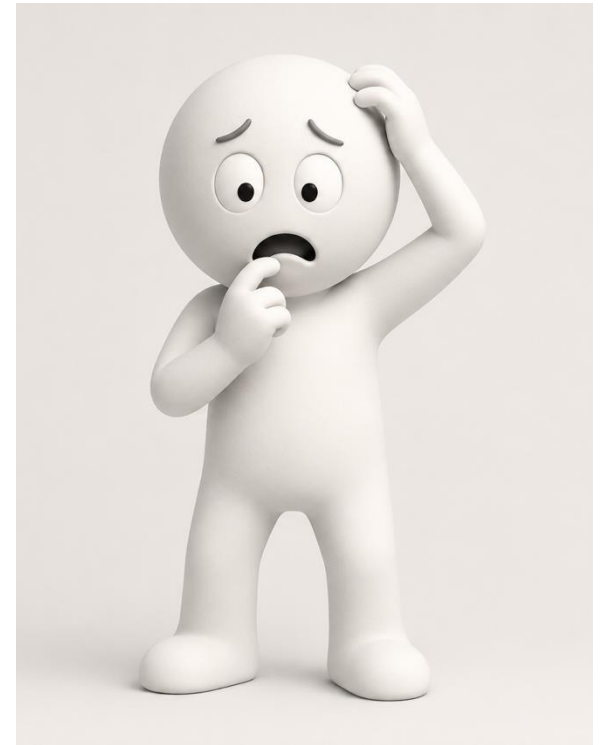
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7. JASP AI
8. Conclusions

Bayesian benefits – Confidence intervals

- What's the definition of a confidence interval?

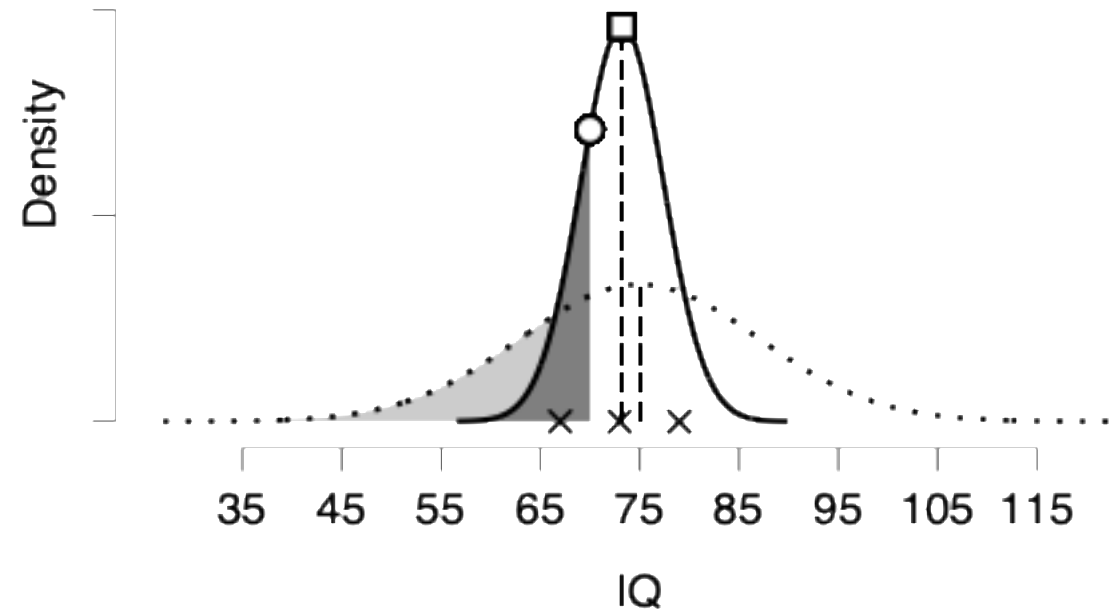
"An X% confidence interval is an interval produced by a procedure that, across repeated sampling, contains the true value of θ in X% of cases (Neyman, 1937)."

- The X% is a property of the procedure that generated the interval, not of any single interval it produced.
- We cannot say how likely it is that the present interval contains the true value
- Bob, the Florida Killer, had IQ test scores of {73, 67, 79}: 95% CI [58, 88]
- We cannot:
 - Say the interval that contains his true IQ score with 95% probability
 - Say how likely it is his IQ is lower than 70



Bayesian benefits – Posterior distribution

- The posterior provides all the answers we were looking for
- Example: Bob, the Florida Killer, had IQ test scores of {73, 67, 79}
- 95% credible interval: [65.09, 81.39] contains the true value with 95% probability
- The probability Bob's IQ is lower than 70 is 22% given the data and the model



Bayesian benefits – Testing: p-value

- What is the definition of a p-value?

The probability of observing data at least as extreme as the observed, if the null were true.

- Depends on data that was never seen:
 - We have not administered our experiment many times
 - The p-value counts not only the data we observed, but also more extreme data we did not observe
- You can only reject the null, not quantify support for it or the alternative
- Assume we tested Bob's IQ 20 times, and his mean IQ score was 73, $p = .038$:

If Bob's true IQ were exactly 70, we would see a sample mean at least 3 points away from 70 about 3.8% of the time.

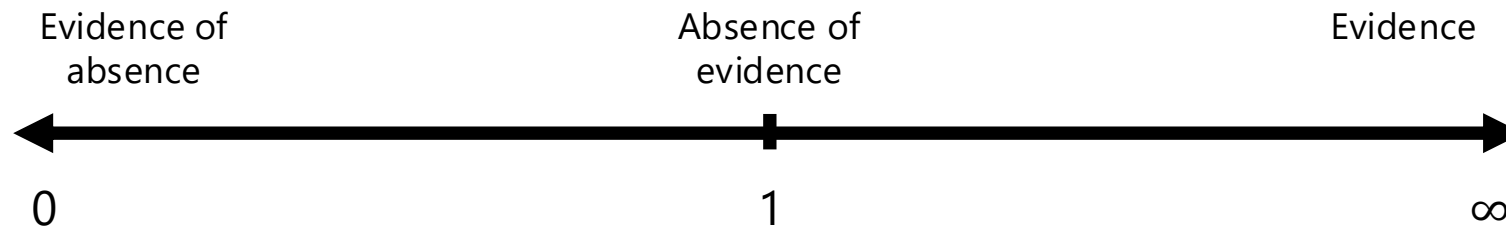


Bayesian benefits – Testing: Bayes factor

- Quantify the absence of evidence for an effect: The corresponding BF_{10} for Bob's IQ equals 1.74:

The BF_{10} of 1.74 is falling into the inconclusive category, so there is neither evidence for nor against an effect.

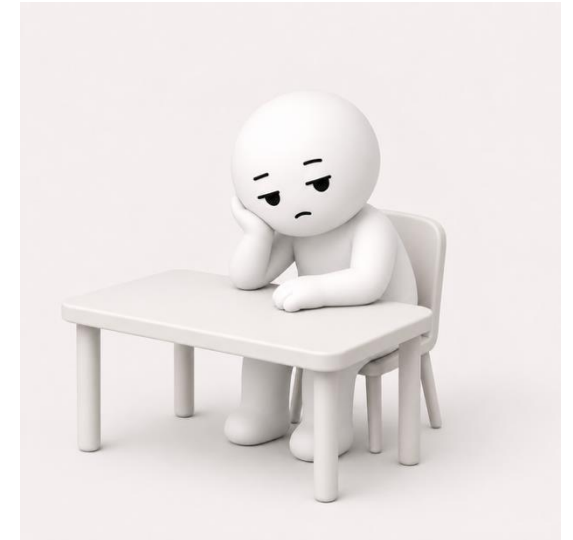
- A p-value of .5 only tells us the data are unsurprising under the null, compatible with evidence for the null and with data that are uninformative
- A BF of 1 tells us that there is neither evidence for the null nor for the alternative



	p-value	Bayes factor
Evidence of absence	No	$BF_{10} > 1$
Absence of evidence	No	$BF_{10} = 1$
Evidence	Quantify evidence against the null	$BF_{10} > 1$

Bayesian benefits – Optional stopping: p-value

- Run an intervention study
- Plan for 60 participants
- Recruitment is slow and each session is expensive
- After 25 participants you run the analysis, $p = .21$
- After 40 participants you run the analysis, $p = .03$, you stop, publish with a significant effect at $n = 40$.
- Why is this wrong?
- A p-value is a statement about how often a procedure produces extreme results when there's no effect. It depends on what that procedure was
- Your procedure was to collect participants until the result was significant. But the p-value was computed as though you had fixed $n = 40$ in advance and tested once. It describes a study you did not run, so the 3% does not apply to what you did.



Bayesian benefits – Optional stopping: Bayes factor

- Run an intervention study
- Plan for 60 participants
- Recruitment is slow and each session is expensive
- After 25 participants you run the analysis, $BF_{10} = 0.6$
- After 40 participants you run the analysis, $BF_{10} = 2.1$
- The BF depends only on $p(\text{data} | H)$, the probability of the data you collected
- You could stop at any time and publish your results
- The evidence can still mislead you: if the null is true, the probability the BF_{10} ever reaches k (e.g., 3) at any point during sampling is at most $1/k$ (e.g., 30%)

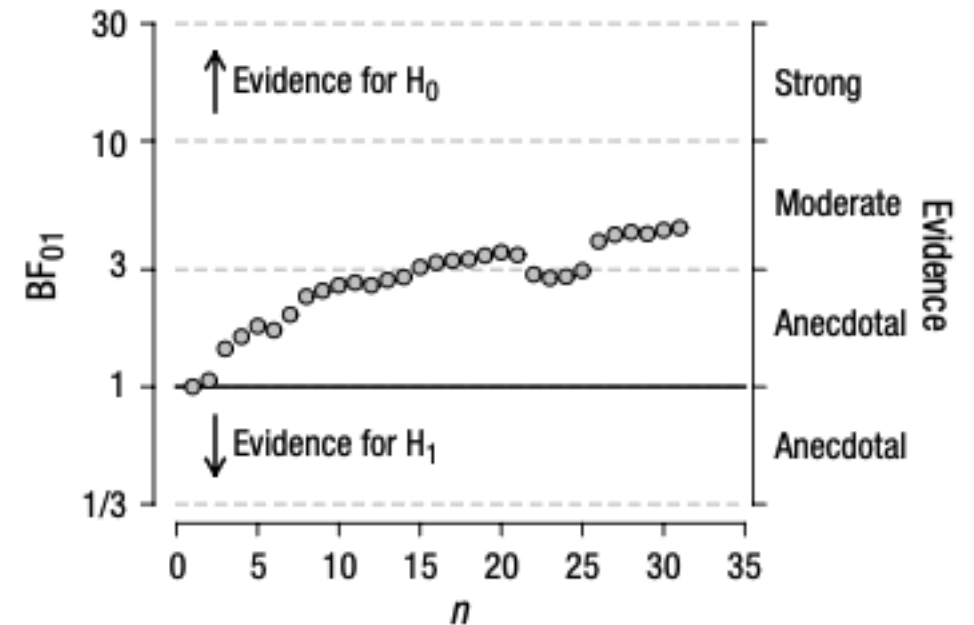
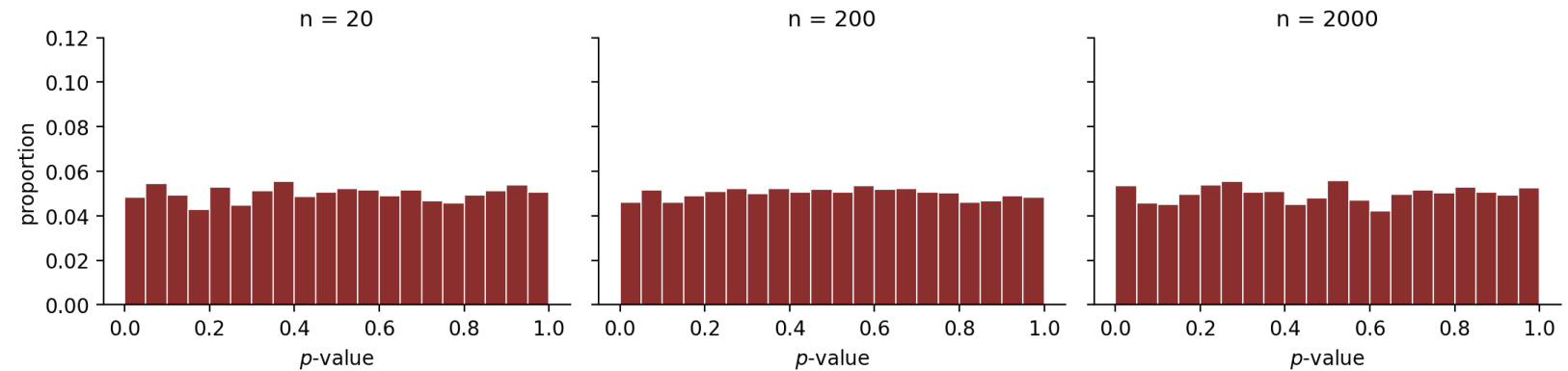


Figure from Wagenmakers, Morey & Lee (2016)

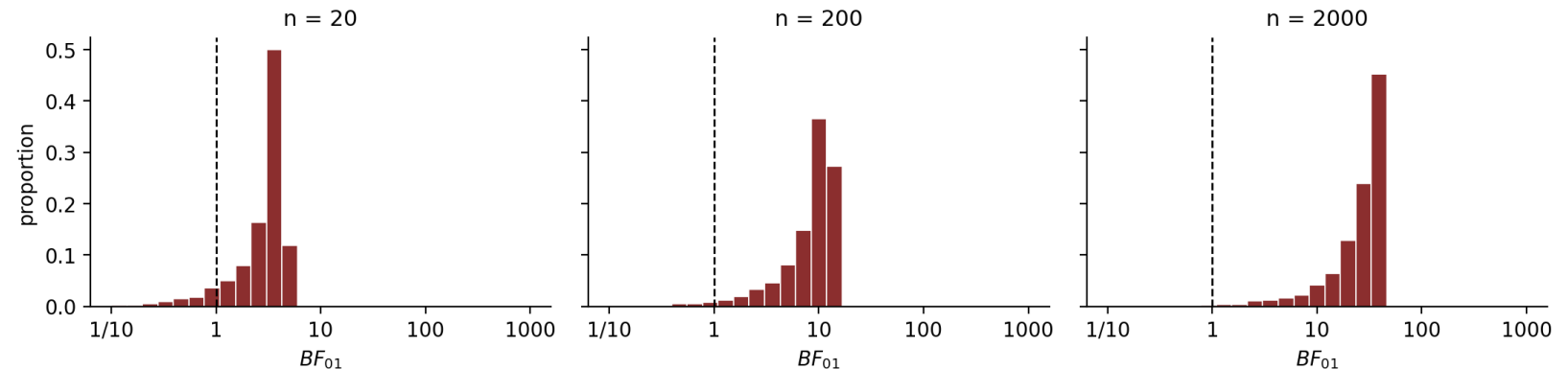
Bayesian benefits – Bayes factor: Consistency

- If H_1 is true and n grows:
 - p -value goes to 0
 - BF_{10} goes to infinity
- If H_0 is true and n grows:
 - p -value is uniformly distributed $[0, 1]$
 - BF_{01} goes to infinity

p -values under H_0 : uniform at every sample size

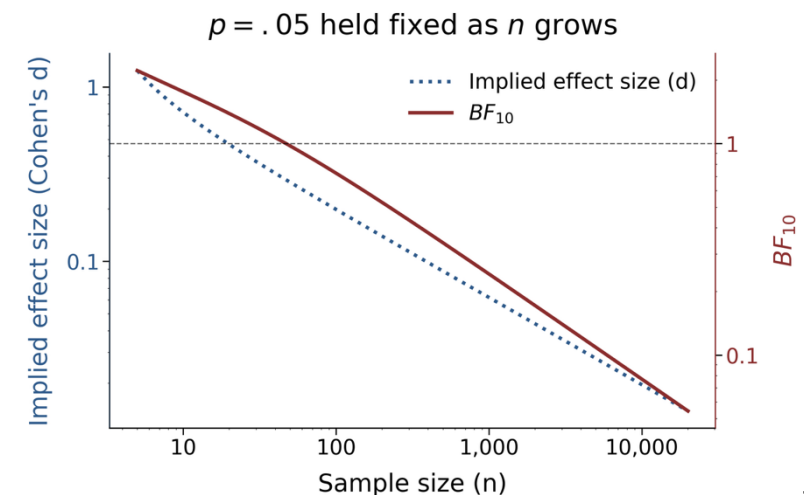
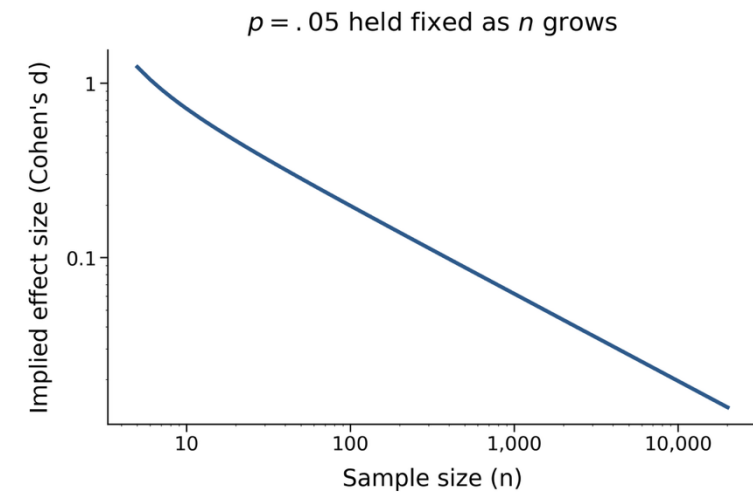


Bayes factors under H_0 : evidence for H_0 grows with sample size



Bayesian benefits – Jeffreys-Lindley paradox

- Assume we conduct studies with growing n , and we fix the p-value at $p = .05$, the effect size shrinks towards 0
- At $n = 20$, the p-value flags a substantial effect
- At $n = 10000$, the effect is tiny
- The p-value stays exactly the same and the decision rule gives the exact same verdict
- The p-value tells you neither how large the effect is nor how strong the evidence
- With larger n also even small effects become significant
- But the BF starts favoring the null



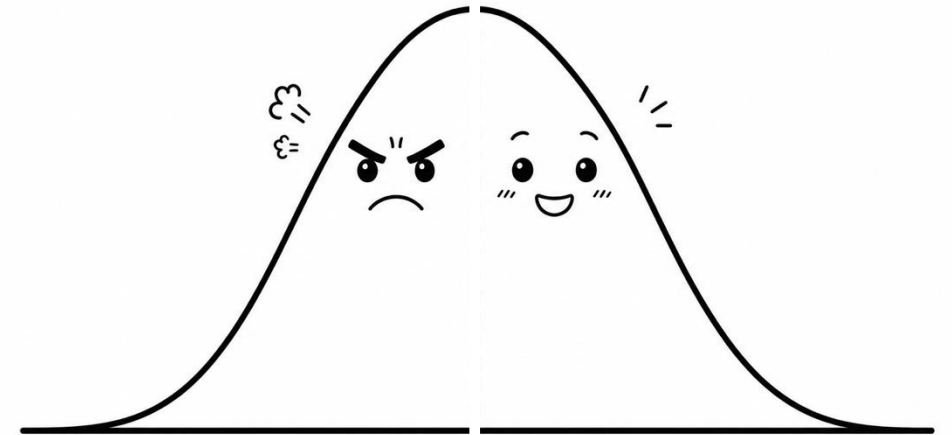
Bayesian benefits – Bayes factors: Testing multiple hypotheses

- Frequentist:
 - Testing is pairwise and sequential
 - Nesting is required
 - Multiplicity correction
- Bayes:
 - Multiple hypotheses can be evaluated jointly: e.g., in regression all subset of predictors evaluated at once
 - Nesting is not required for comparison, any hypotheses that make predictions about the same data can be compared
 - Model averaging: weigh models and inferences with posterior model probability

Bayesian benefits – Prior: A nuisance?

Can you think of arguments why the prior is a nuisance?

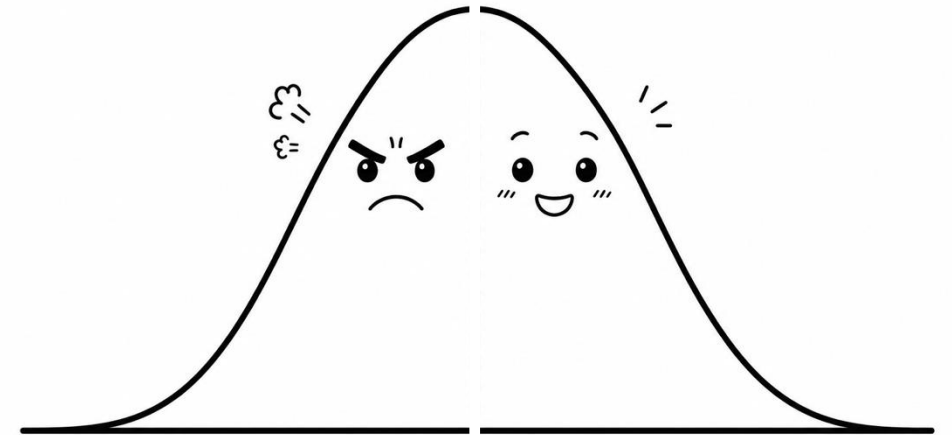
- Subjective: two researchers with different priors reach different conclusions
- Sometimes hard to define
- In testing, more data does not diminish the prior influence
- Prior-hacking:
 - A very informative prior favours the alternative
 - A very wide prior favours the null
- Reviewers usually do not like it



Bayesian benefits – Prior: An asset?

Can you think of some arguments that make the prior an asset?

- It is a key ingredient making Bayesian inference possible
- Prior information is there, why ignore it?
- Assumptions are visible and testable:
 - Prior predictive simulation
 - Prior sensitivity analysis
- A sharp prior rewards if right, and penalizes if wrong
- Small samples: the prior supplies information the data cannot
- Transparency is key



Bayesian benefits – Missing data

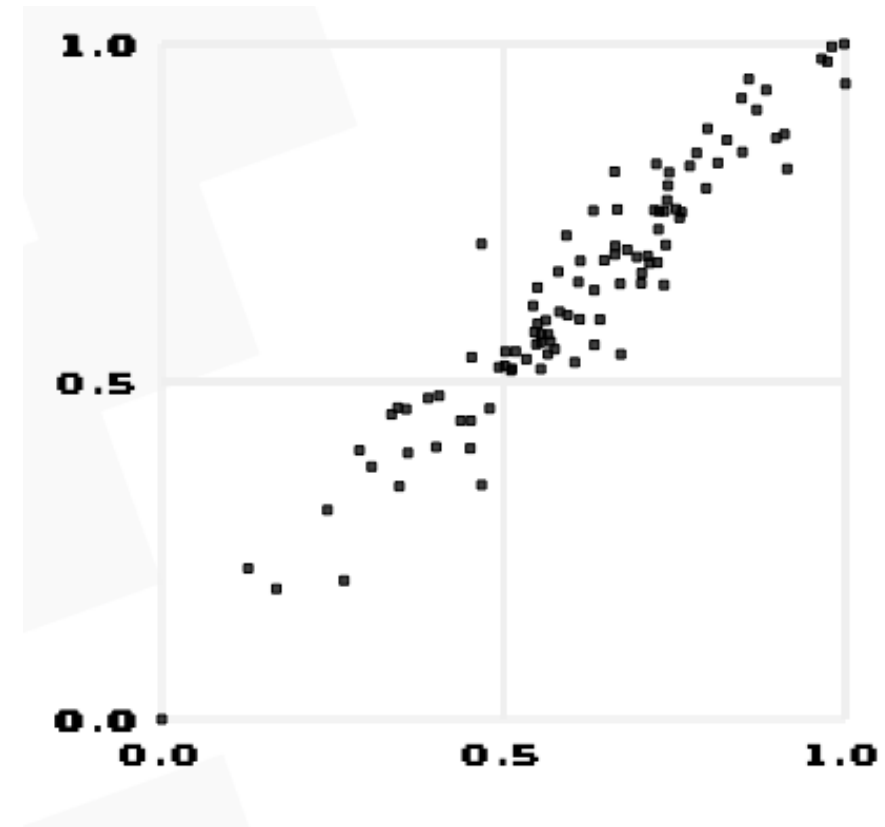
- A missing value is a parameter to be estimated, so it has its own posterior
- Missing values are estimated jointly with the model parameters
- Multiple imputation approximates Bayesian handling of missing values
- MNAR: Prior can be used to make assumptions about the missingness, varying the prior shows how much conclusions depend on it
- However: JASP is incomplete in its capabilities to account for missing data; Bayesian R-packages often handle this much better



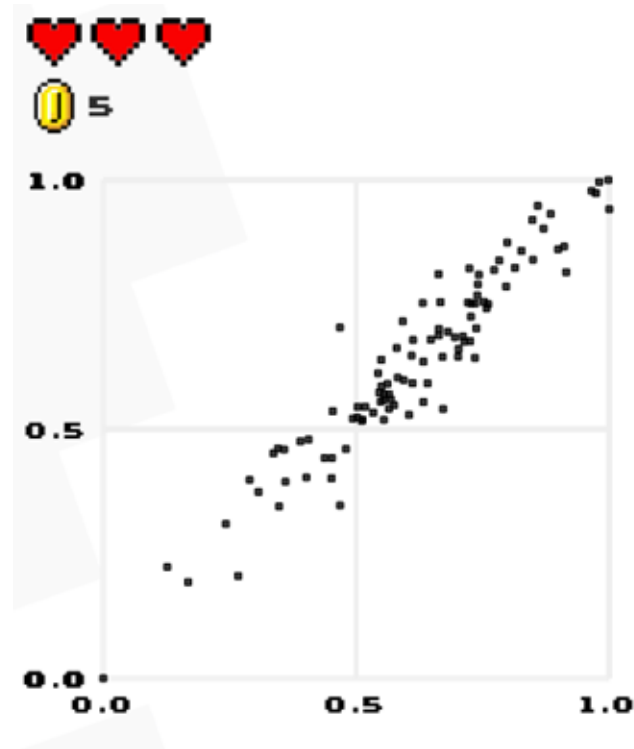
Contents – Day 1

1. Introduction
2. Bayesian basics
3. Bayesian estimation
4. Bayesian testing
5. Bayesian benefits
- 6. JASP with real data**
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Bayesian Correlation



Correlation



HIGH SCORE 0 **MAIN MENU**

NEXT

TRUE R	0.95
GUESSED R	0.93
DIFFERENCE	0.02
STREAKS	1
MEAN ERROR	0.02

♥ +1 **🪙 +5**

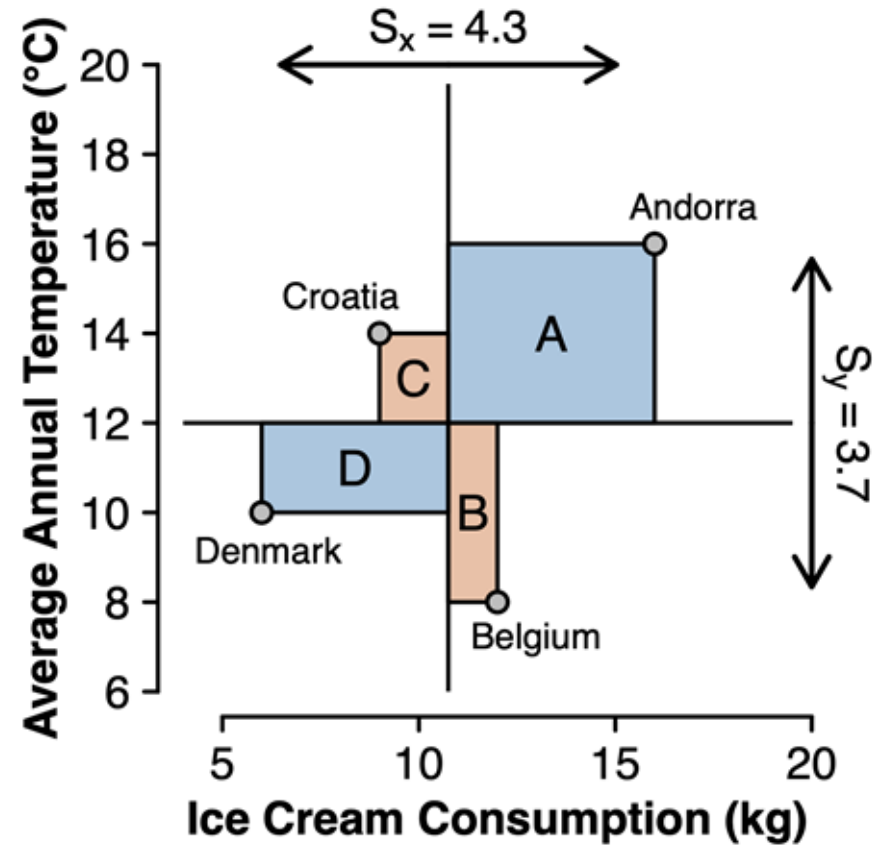
<https://guessthecorrelation.com/>

Correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}},$$

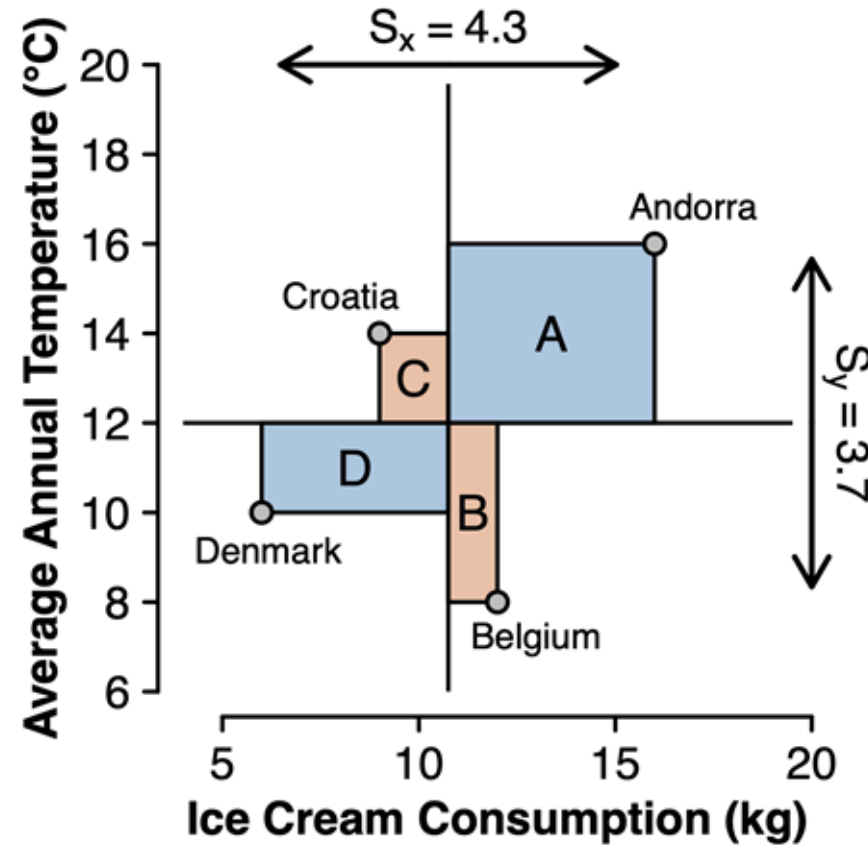
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... is the difference between concordant and discordant areas, normalized by the total variation

Correlation

- Null hypothesis: The correlation is 0.
“There is no association between two continuous variables”

Correlation

- Null hypothesis: The correlation is 0.
“There is no association between two continuous variables”
- Assumptions: Bivariate normality
 - non-parametric alternatives, for instance for ranks

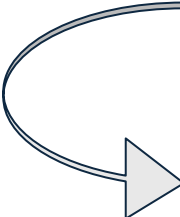
Classical Correlation hypothesis test

$$H_0: \rho = 0 ; H_1: \rho \neq 0$$

Classical Correlation hypothesis test

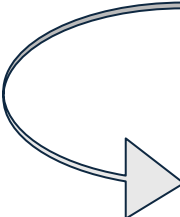
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r_{xy} is the only thing the sample contributes (the observed values collapsed in a single number). Everything after this is theory about r , i.e., about the reference distribution of r when H_0 is true.

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...centered at 0 and symmetric, it's tight for large n , wide for small n . Its shape is derived under the assumption that H_0 is true.

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...now we pretend ρ is truly 0, and ask how r would vary across hypothetical repeated samples.

...if ρ really were 0, probability of seeing an r at least this extreme just from sampling noise. Compare p to α , **reject or fail to reject.**

...centered at 0 and symmetric, it's tight for large n , wide for small n . Its shape is derived under the assumption that H_0 is true.

Classical Correlation hypothesis test

$$H_0: \rho = 0$$

...frequentist: ρ is a fixed unknown constant; the data (r) are random. The probability statement is about data-given-a-fixed-null.

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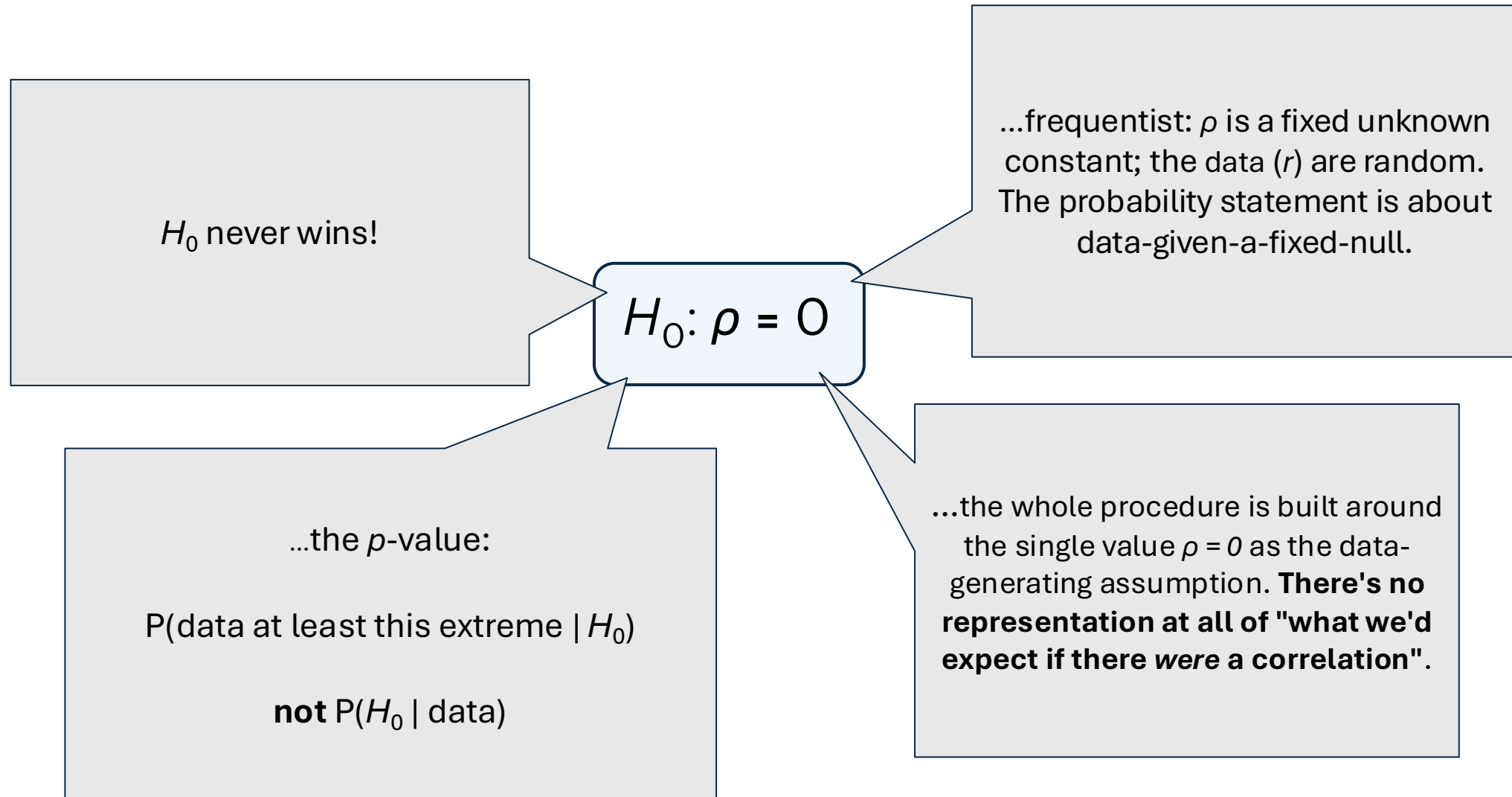
...the p -value:

$P(\text{data at least this extreme} \mid H_0)$

not $P(H_0 \mid \text{data})$

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Classical Correlation hypothesis test



Bayesian Correlation - Theory

$$H_0: \rho = 0$$



$$H_1: \rho \neq 0$$

Bayesian Correlation - Theory

$H_0: \rho = 0$

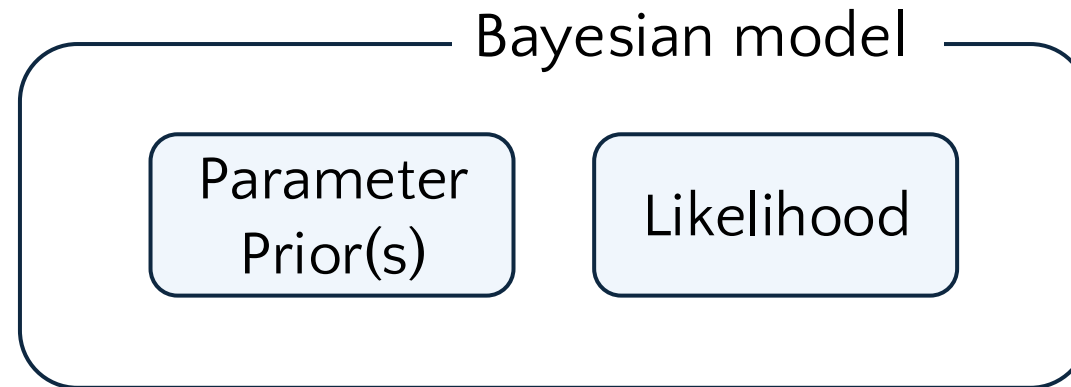


$H_1: \rho \neq 0$

$$\underbrace{\frac{P(H_1 | d)}{P(H_0 | d)}}_{\text{posterior odds}} = \underbrace{\frac{P(d | H_1)}{P(d | H_0)}}_{\text{Bayes factor}} \times \underbrace{\frac{P(H_1)}{P(H_0)}}_{\text{prior odds}}.$$

= evidence

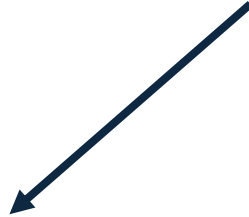
Bayesian Correlation - Theory



Bayesian Correlation - Theory

Likelihood

Density: $f(\text{data} \mid \text{parameters})$



...fix the parameters (say $\mu = 0$, $\sigma = 1$) and let the data vary: now it tells you how probable different data outcomes are. That's a **probability distribution**.

Bayesian Correlation - Theory

Likelihood

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...fix the data (the study is done, those numbers are locked) and try out different parameter values. The **likelihood** scores each one: how well does this set of parameters explain the data actually observed?
Higher = better fit.

Bayesian Correlation - Theory

Likelihood

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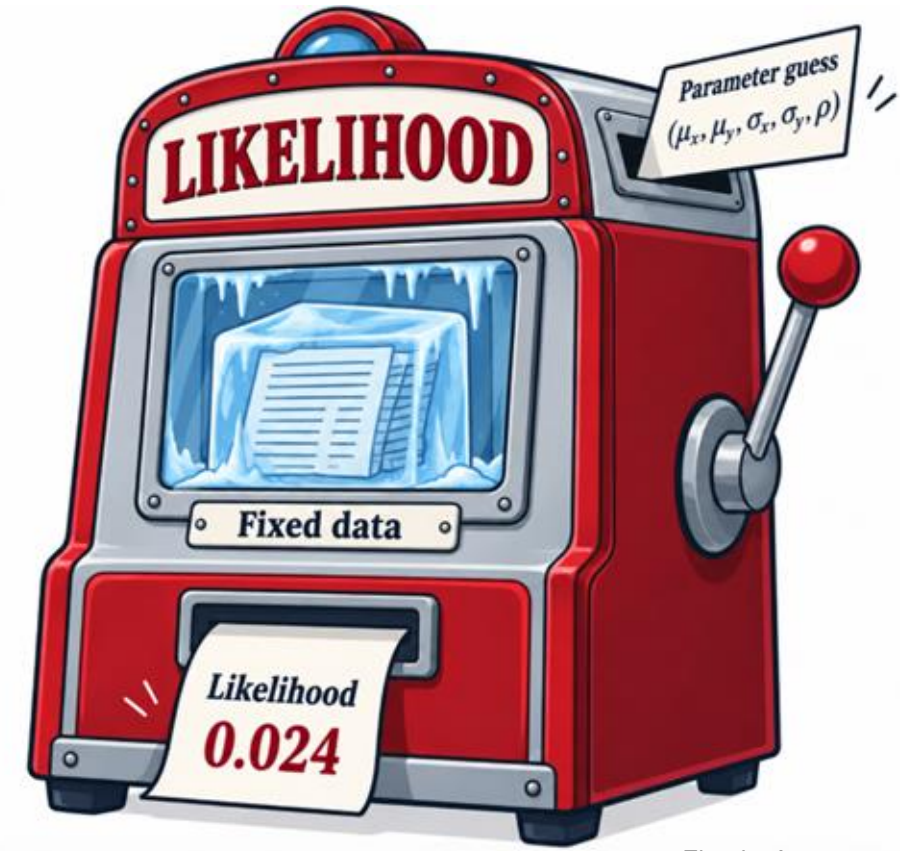
...how good is $\rho = 0.3$?
how good is $\rho = 0.7$?
how good is $\rho = 0$?



Bayesian Correlation - Theory

Likelihood

The *likelihood function* is just the collection of all those output numbers as you vary the guess

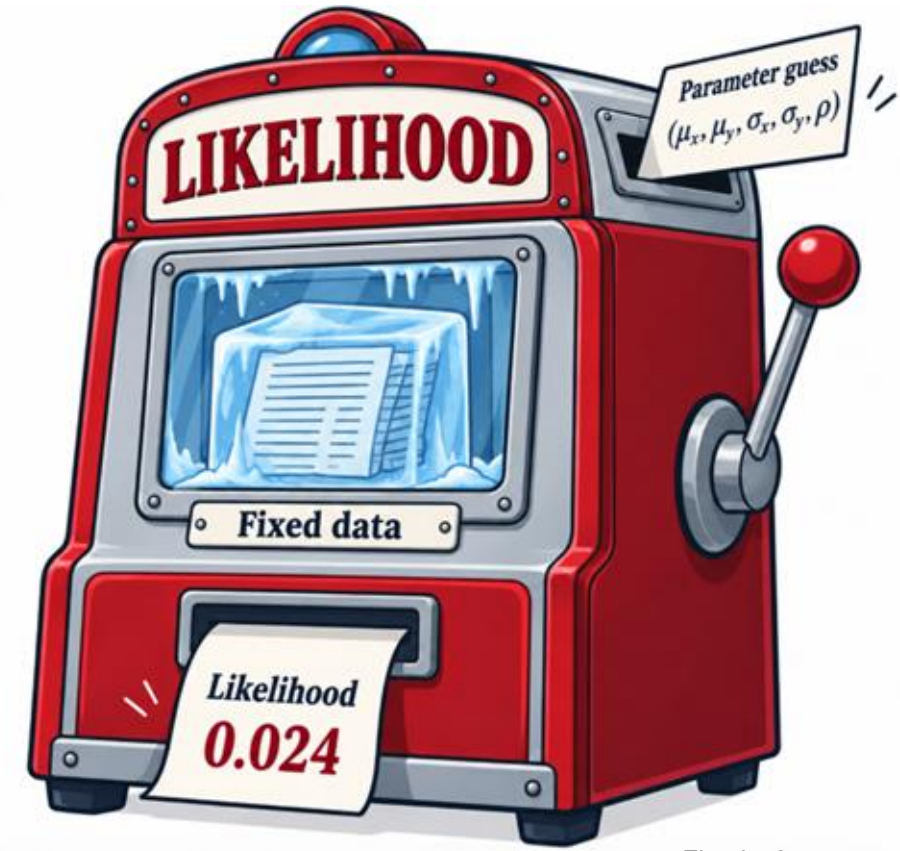


Thanks Astra

Bayesian Correlation - Theory

Likelihood

$$\underbrace{L(\theta_0 \mid d)}_{\text{one guess} \rightarrow \text{one score}}$$



Thanks Astra

Bayesian Correlation - Theory

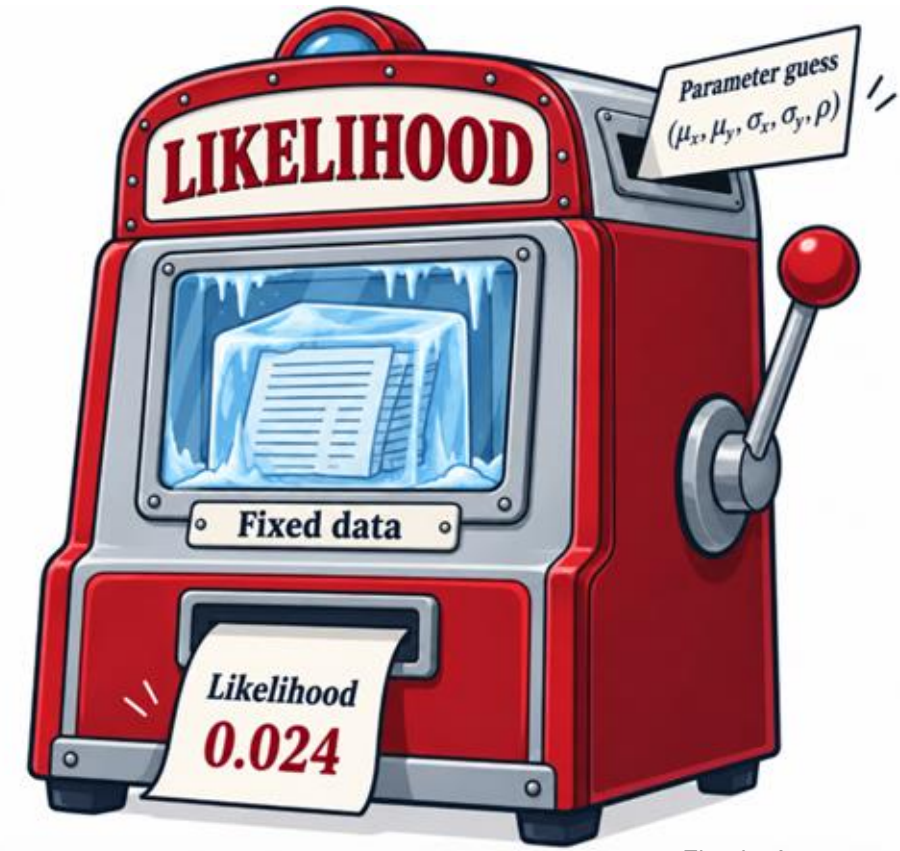
Likelihood

$$\underbrace{L(\theta_0 \mid d)}$$

one guess → one score

$$\underbrace{L(\theta \mid d) \text{ for all } \theta}$$

every guess → a surface of scores



Thanks Astra

Bayesian Correlation - Theory

Likelihood

$$\underbrace{L(\theta_0 \mid d)}$$

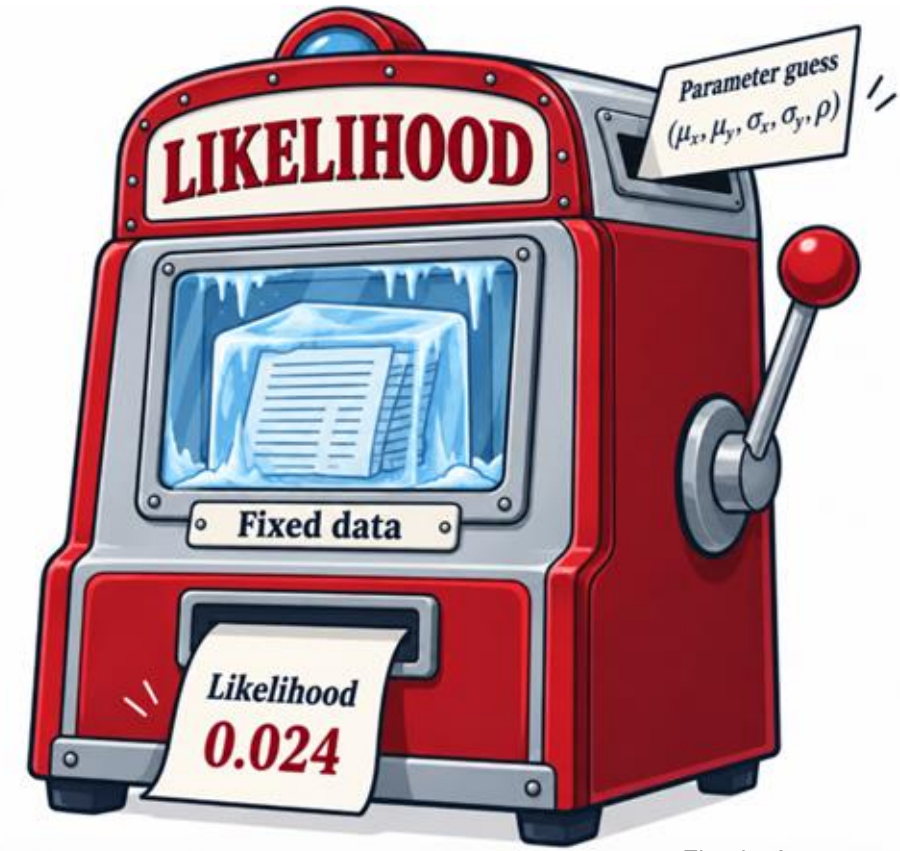
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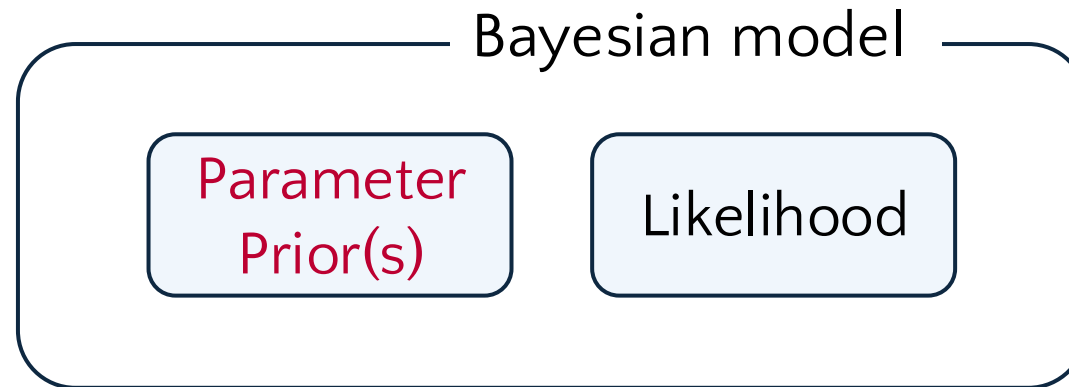
$$\underbrace{P(d \mid H) = \int L(\theta \mid d) p(\theta \mid H) d\theta}$$

prior-weighted average → one number for the model



Thanks Astra

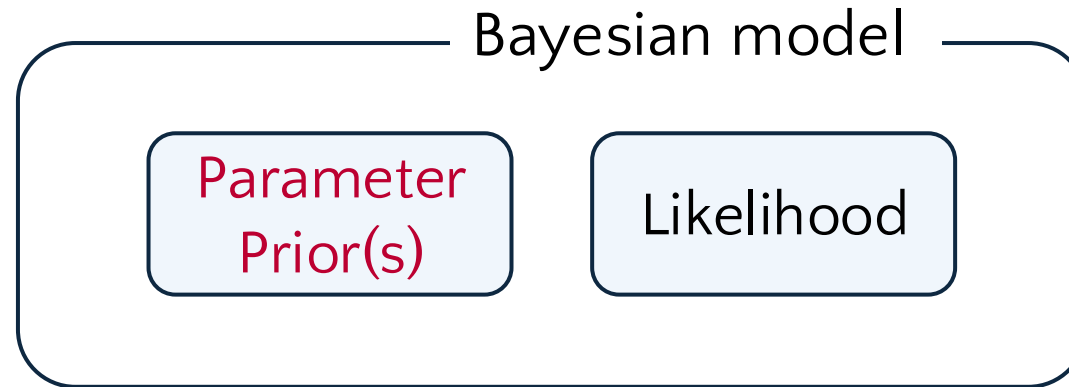
Bayesian Correlation - Theory



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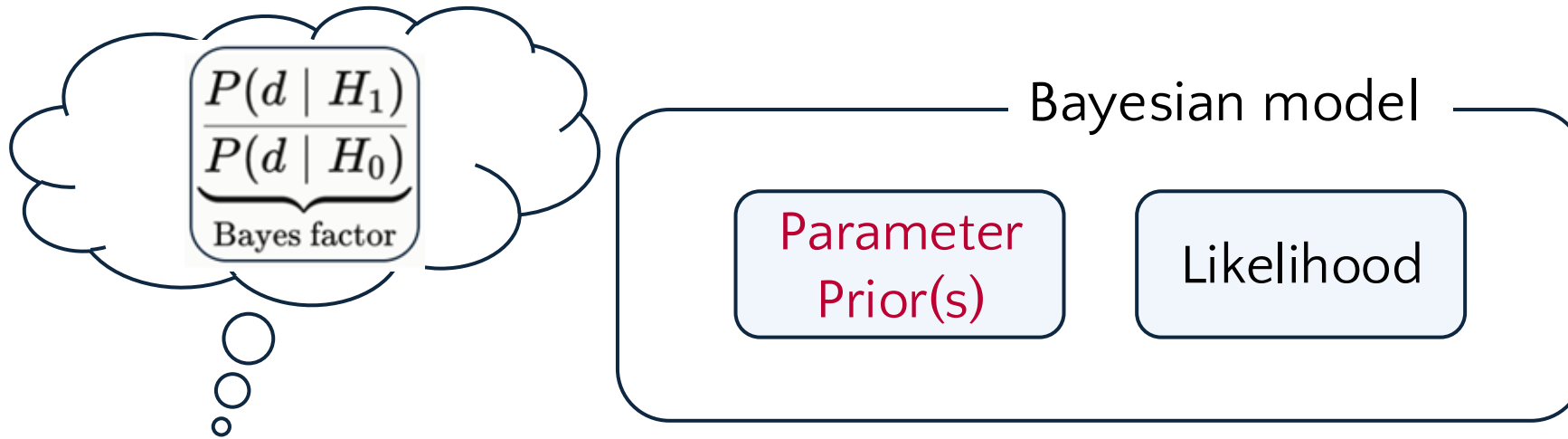
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Bayesian Correlation - Theory



$$\underbrace{P(d \mid H) = \int L(\theta \mid d) p(\theta \mid H) d\theta}_{\text{prior-weighted average} \rightarrow \text{one number for the model}}$$

Bayesian Correlation - Theory



$$P(d | H) = \int L(\theta | d) p(\theta | H) d\theta$$

prior-weighted average → one number for the model

Bayesian Correlation - Theory

Prior(s)

Bayesian Correlation - Theory

Prior(s)

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

Bayesian Correlation - Theory

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$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

$$P(d | H_1) = \int \int \int \int \int L(\dots, \rho) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation - Theory

Prior(s)

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$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

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$$P(d \mid H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

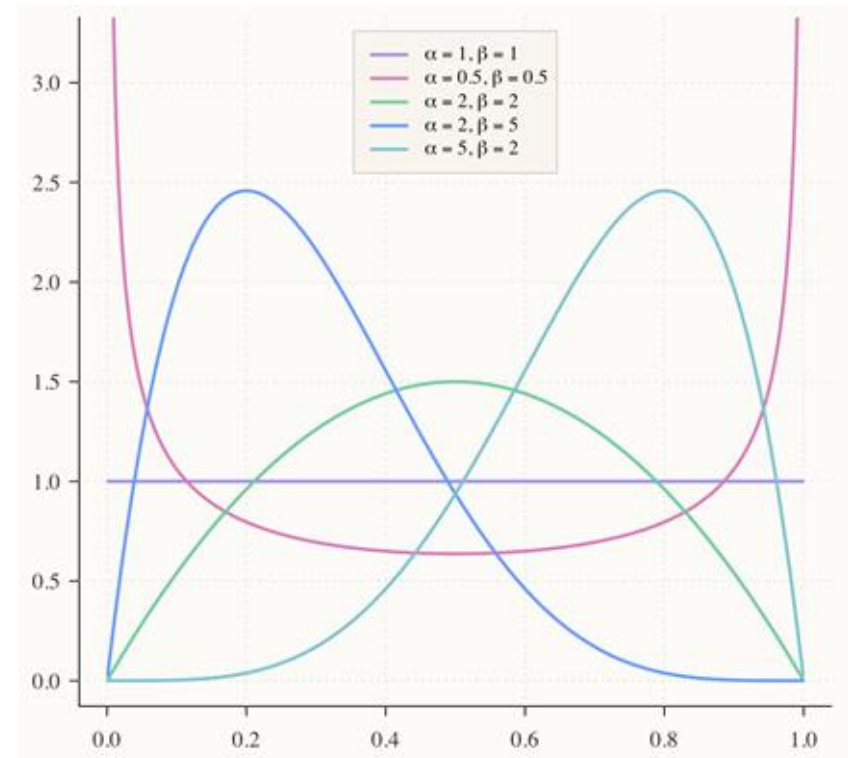
$$P(d \mid H_1) = \int \int \int \int \int L(\dots, \rho) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y}^{\frac{1}{2}} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation - Theory

Prior(s)

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Bayesian Correlation - Theory

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$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$

$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_0) = \int \int \int \int L(\dots, \rho=0) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} d\mu_x d\mu_y d\sigma_x d\sigma_y$$

$$P(d | H_1) = \int \int \int \int \int L(\text{--- } \alpha=2, \beta=2 \text{ ---}) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

Bayesian Correlation - Theory

$$H_0: \rho = 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y$$



$$H_1: \rho \neq 0$$

$$\mu_x, \sigma_x, \mu_y, \sigma_y, \rho$$

$$P(d | H_1) = \int \int \int \int \int L(\text{--- } \alpha=2, \beta=2 \text{ ---}) \cdot 1 \cdot 1 \cdot \frac{1}{\sigma_x} \frac{1}{\sigma_y} \frac{1}{2} d\mu_x d\mu_y d\sigma_x d\sigma_y d\rho$$

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Bayesian Correlation - Theory

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$$\frac{P(d | H_1)}{P(d | H_0)} = \underbrace{\text{Bayes factor}}_{\text{= evidence}} = \text{BF}_{10}$$

Bayesian Correlation - Theory

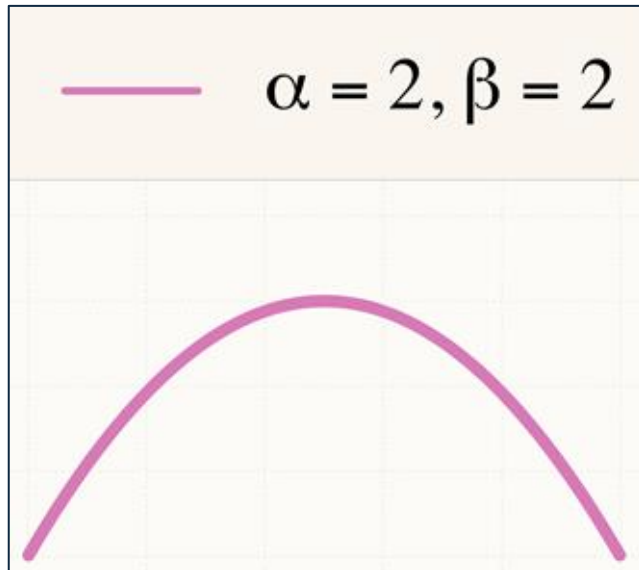
$$\underbrace{\frac{P(d \mid H_1)}{P(d \mid H_0)}}_{\text{Bayes factor}} = \text{BF}_{10}$$

= evidence

Testing (Bayes factor): "Is there an effect at all?":

- compares $H_0: \rho = 0$ against $H_1: \rho \neq 0$.
- Output: a number weighing one model against the other.

Bayesian Correlation - Theory



Estimation (posterior): "Assuming there is an effect, how big is it, and how sure are we?"

- entirely based on H_1
- Output: a whole updated distribution over ρ

Bayesian Correlation - Theory

$$\underbrace{p(\theta \mid d)}_{\text{posterior}} = \frac{\overbrace{L(\theta \mid d)}^{\text{likelihood}} \overbrace{p(\theta \mid H_1)}^{\text{prior}}}{\underbrace{\int L(\theta \mid d) p(\theta \mid H_1) d\theta}_{\text{normalizing constant} = P(d|H_1)}}$$



Bayesian Correlation - Theory

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Bayesian Correlation - Theory

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$$\underbrace{p(\rho \mid d)}_{\text{posterior for the effect}} = \int \underbrace{p(\theta \mid d)}_{\text{joint posterior}} d\eta \quad \text{where } \eta = (\mu_x, \mu_y, \sigma_x, \sigma_y)$$

Bayesian Correlation - Example Part 1



Let's do a **Bayesian Correlation analysis** together.

Example data set:

Adam Sandler (JASP data library)

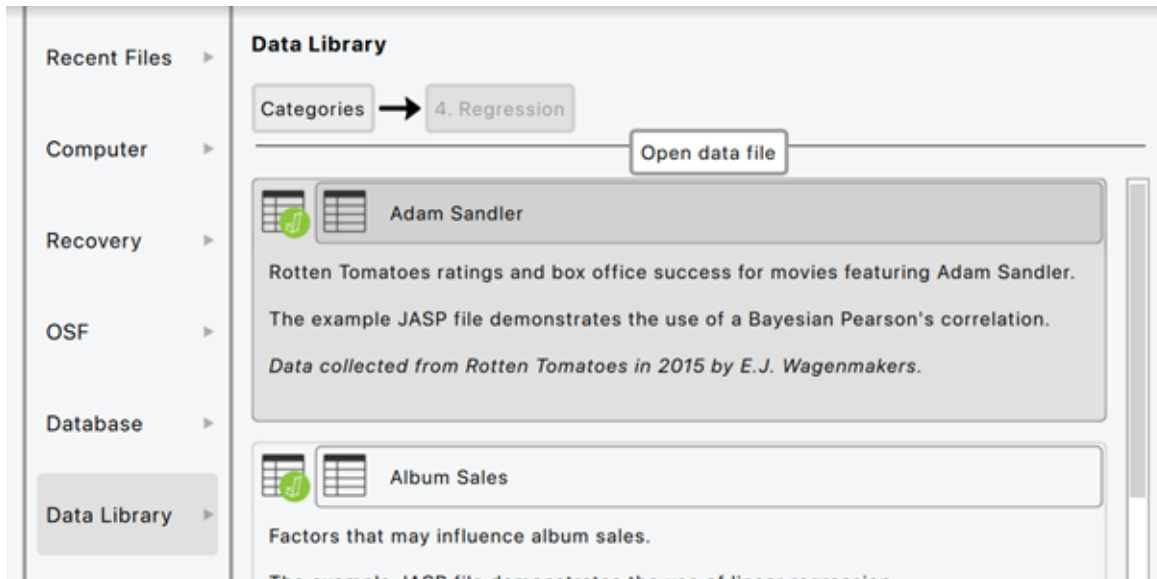
Bayesian Correlation - Example Part 1



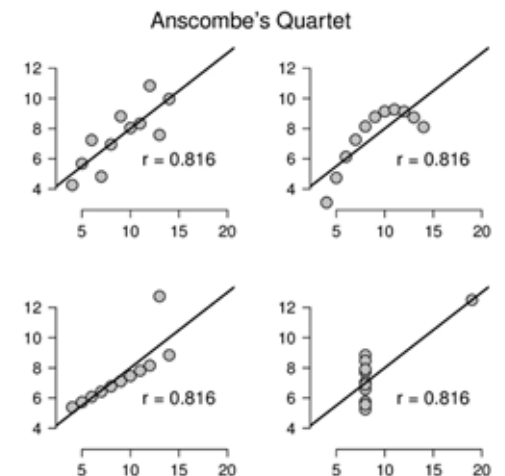
Let's do a **Bayesian Correlation analysis** together.

Example data set:

Adam Sandler (JASP data library)



P.S.: Always plot your data!



Bayesian Correlation - Exercise



Your turn!

Exercise data set: **Do lonely people take hotter showers?**

use showers.csv

Feel free to first do a classical analysis.
Then, please do a Bayesian reanalysis!

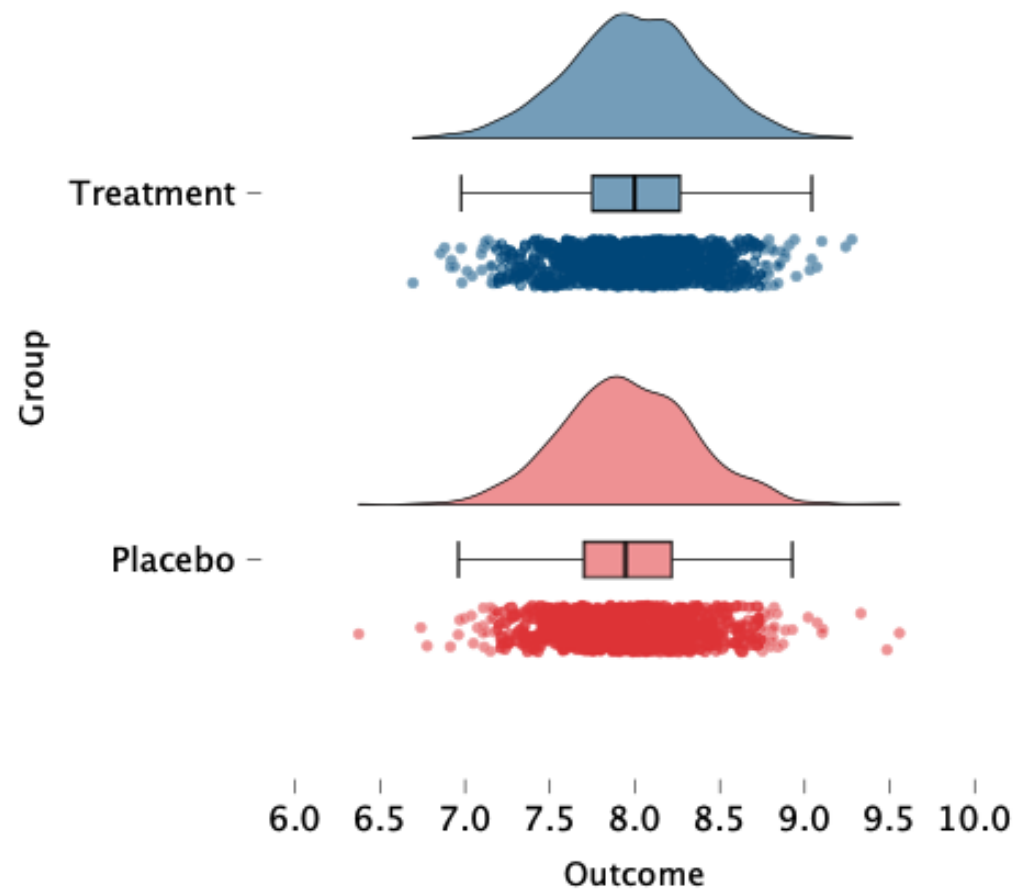


Bayesian Correlation - Further Reading

Wagenmakers, E. J., Godmann, H., De Ruiter, J., & van Doorn, J. (2023). Visualizing the Equation for the Sample Correlation Coefficient.

Wagenmakers, E. J., Verhagen, J., & Ly, A. (2016). How to quantify the evidence for the absence of a correlation. *Behavior Research Methods*, 48(2), 413-426. !

Bayesian *t*-Test



t -Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure?

t -Test

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t -Test

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- Others:
 - One-sample t -test
 - Paired t -test

t -Test

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- Secondary question: If there is an effect, how large is it?

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 - Sample effect size -> Cohen's d (difference in means / common sd)

t-Test

- Most commonly: Is there a difference between two groups on a continuous outcome measure? -> **Independent samples *t*-test**
- Secondary question: If there is an effect, how large is it?
 - Sample effect size -> Cohen's *d* (difference in means / common sd)
 - Population version: $\delta = \mu / \sigma$

Classical t -Test - Theory

group 1 $\sim \mathcal{N}(\mu_1, \sigma^2)$, group 2 $\sim \mathcal{N}(\mu_2, \sigma^2)$.

$\delta = \mu_1 - \mu_2$, $H_0 : \delta = 0$ vs $H_1 : \delta \neq 0$.

- Assumptions:
 - Both groups drawn from independent Normal distributions
 - Shared variance (other exist, i.e., Welch's t -test)

Classical t -Test - Theory

$$\bar{x}_1 - \bar{x}_2$$

$$t = \frac{\text{signal}}{\text{noise}} = \frac{\text{observed gap between groups}}{\text{how much that gap would wobble by chance}}$$

$$s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Two-sided ($H_1 : \delta \neq 0$): both tails,
 $p = P(|T_\nu| \geq |t_{\text{obs}}|) = 2 P(T_\nu \geq |t_{\text{obs}}|)$

Classical t -Test - Theory

- p is P (data at least this extreme | H_0), **not** P (H_0 | data)

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Classical t -Test - Theory

- p is $P(\text{data at least this extreme} \mid H_0)$, **not** $P(H_0 \mid \text{data})$
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- The test does not quantify evidence: A tiny, trivial gap becomes significant with enough observations

Two-sided ($H_1 : \delta \neq 0$): both tails,

$$p = P(|T_\nu| \geq |t_{\text{obs}}|) = 2 P(T_\nu \geq |t_{\text{obs}}|)$$

Bayesian t -Test - Theory

Model

$$\text{group 1} \sim \mathcal{N}(\mu_1, \sigma^2), \quad \text{group 2} \sim \mathcal{N}(\mu_2, \sigma^2).$$

$$\delta = \mu_1 - \mu_2, \quad H_0 : \delta = 0 \quad \text{vs} \quad H_1 : \delta \neq 0.$$

Bayesian t -Test - Theory

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Bayesian t -Test - Theory

Model

$$x_1 \sim \mathcal{N}\left(\mu + \frac{\alpha}{2}, \sigma^2\right)$$

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Bayesian t -Test - Theory

Prior(s)

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The test-irrelevant parameters μ and σ are common to all of the models. These parameters can be assigned prior distributions that are *weakly informed by the observed data...*

(important to remember for hands-on modeling: variance parameters cancel when they only set the scale of the noise; they distort the effect when they also decide how much of the data counts as noise)

Bayesian t -Test - Theory

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Bayesian t -Test - Theory

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- Pick the prior (or family of priors) that meets a series of desiderata

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- Earlier experiments on similar topics may give more specific information
- Expert knowledge yields relatively precise predictions
- Drawback: effortful and “subjective”

Objective

- Pick the prior (or family of priors) that meets a series of desiderata
- Advantage: can be applied in any scenario, forms a point of reference for a more detailed analysis, “objective”

Bayesian t -Test - Theory

Prior(s)

Objective – desiderata

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Bayesian t -Test - Theory

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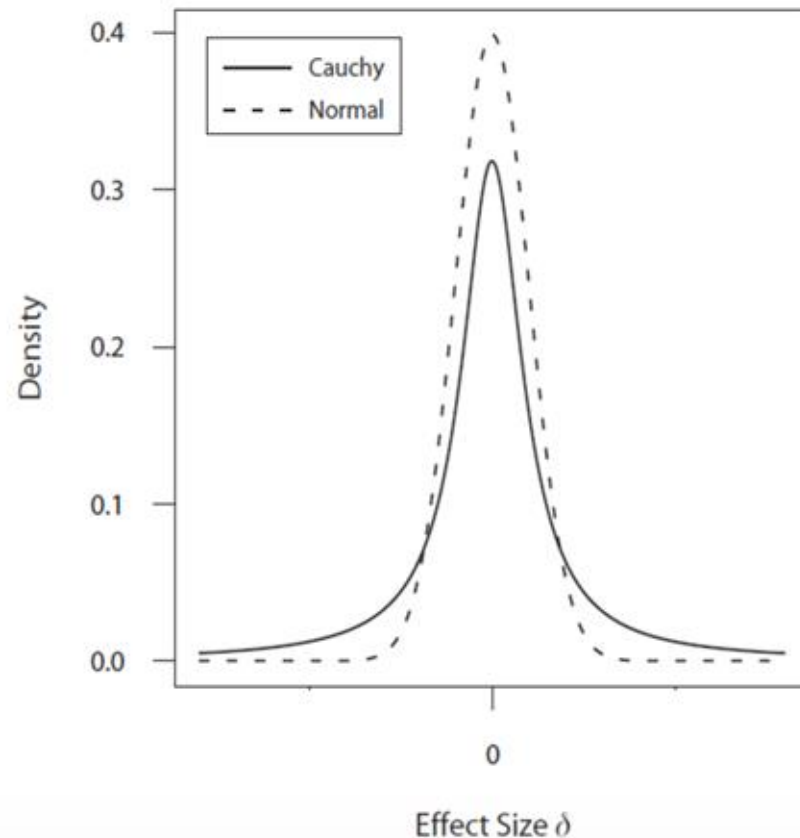
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Candidates

- Normal?
- Uniform?
- Cauchy!

Bayesian t -Test - Theory

Prior(s)



Candidates

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- Uniform?
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Bayesian t -Test - Theory

$$H_0 : \delta = 0 \quad H_1 : \delta \sim \text{Cauchy}\left(0, \frac{1}{\sqrt{2}}\right)$$

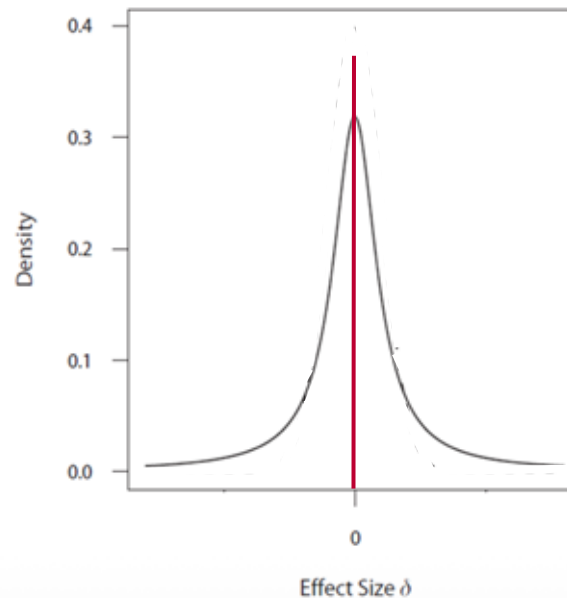
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$$\delta = \frac{\alpha}{\sigma}$$

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Bayesian t -Test - Theory

Testing:

$$\underbrace{\frac{P(d \mid H_1)}{P(d \mid H_0)}}_{\text{Bayes factor}}$$

= evidence

Estimation:

$$\underbrace{p(\delta \mid d)}_{\text{posterior for the effect}} = \int \underbrace{p(\theta \mid d)}_{\text{joint posterior}} d\eta \quad \text{where } \eta = (\mu, \sigma)$$

Bayesian t -Test - Example



Let's do a **Bayesian t -test analysis** together.

Example data set:

Turning the Wheels of Time

-> JASP data library
(Kitchen Rolls under t -test)



Topolinski, S., & Sparenberg, P. (2012). Turning the hands of time: Clockwise movements increase preference for novelty. *Social Psychological and Personality Science*, 3(3), 308-314.

Bayesian t -Test - Exercise



Your turn!

Exercise data set:

use `insomnia_data.csv`

Feel free to first do a classical analysis.
Then, please do a Bayesian reanalysis :)



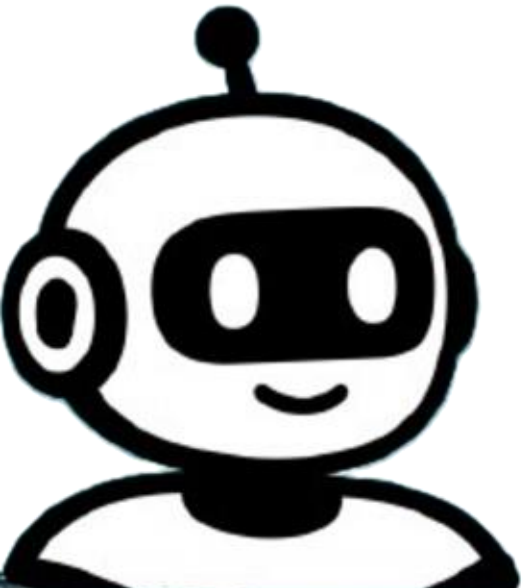
Bayesian t -Test - Exercise



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Bayesian t -Test - Further Reading

Gronau, Q. F., Ly, A., & Wagenmakers, E. J. (2020).
Informed Bayesian t -tests. *The American Statistician*.

Contents – Day 1

1. Introduction
2. Bayesian basics
3. Bayesian estimation
4. Bayesian testing
5. Bayesian benefits
6. JASP with real data
7. JASP AI
- 8. Conclusions**

Further reading

- Lambert, B. (2018). *A student's guide to Bayesian statistics*. SAGE Publications.
<https://sites.math.rutgers.edu/~zeilberg/EM20/Lambert.pdf>
- Wagenmakers, E.-J., Marsman, M., Jamil, T., Ly, A., Verhagen, A. J., Love, J., Selker, R., Gronau, Q. F., Šmíra, M., Epskamp, S., Matzke, D., Rouder, J. N., Morey, R. D. (2018). [Bayesian inference for psychology. Part I: Theoretical advantages and practical ramifications](#). *Psychonomic Bulletin & Review*, 25, 35-57.
- Wagenmakers, E.-J., Morey, R. D., & Lee, M. D. (2016). Bayesian benefits for the pragmatic researcher. *Current Directions in Psychological Science*, 25(3), 169–176.
<https://doi.org/10.1177/0963721416643289>
- Wagenmakers, E.-J., Love, J., Marsman, M., Jamil, T., Ly, A., Verhagen, A. J., Selker, R., Gronau, Q. F., Dropmann, D., Boutin, B., Meerhoff, F., Knight, P., Raj, A., van Kesteren, E.-J., van Doorn, J., Šmíra, M., Epskamp, S., Etz, A., Matzke, D., de Jong, T., van den Bergh, D., Sarafoglou, A., Steingroever, H., Derks, K., Rouder, J. N., & Morey, R. D. (2018). [Bayesian inference for psychology. Part II: Example applications with JASP](#). *Psychonomic Bulletin & Review*, 25, 58-76.

Outlook Day 2

- Multi-model inference
- Mixed models with *brms*: Using R in Rstudio
- How to conduct and report a Bayesian analysis
- Bring your own data
- Beyond the workshop
- Conclusions

Discussion & Feedback

- Do you already see yourself applying Bayesian statistics in your own research?
- If so, how?
- If not, why?
- What do you struggle with?

References

van Doorn, J., van den Bergh, D., Böhm, U., Dablander, F., Derks, K., Draws, T., Etz, A., Evans, N. J., Gronau, Q. F., Haaf, J. M., Hinne, M., Kucharský, Š., Ly, A., Marsman, M., ... Wagenmakers, E.-J. (2021). *The JASP guidelines for conducting and reporting a Bayesian analysis*. *Psychonomic Bulletin & Review*, 28, 813–826. <https://doi.org/10.3758/s13423-020-01798-5>

Mulder, J., Friel, N., & Leifeld, P. (2024). Bayesian testing of scientific expectations under exponential random graph models. *Social Networks*, 78, 40–53. <https://doi.org/10.1016/j.socnet.2023.11.004>

Wagenmakers, E.-J., Morey, R. D., & Lee, M. D. (2016). Bayesian benefits for the pragmatic researcher. *Current Directions in Psychological Science*, 25(3), 169–176. <https://doi.org/10.1177/0963721416643289>